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Walker

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(54) **ESD PROTECTION FOR INTEGRATED
CIRCUIT DEVICES**

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H10D 89/60 (2025.01)

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89/60–931

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,760,444 A * 6/1998 Okumura H01L 27/1203
257/350
6,535,368 B2 3/2003 Andresen
(Continued)

FOREIGN PATENT DOCUMENTS

CN 106024793 B 2/2019
WO 2018190881 A1 10/2018

OTHER PUBLICATIONS

U.S. Appl. No. 17/030,662, Final Office Action, dated Jun. 23, 2023.
(Continued)

Primary Examiner — Jared Fureman

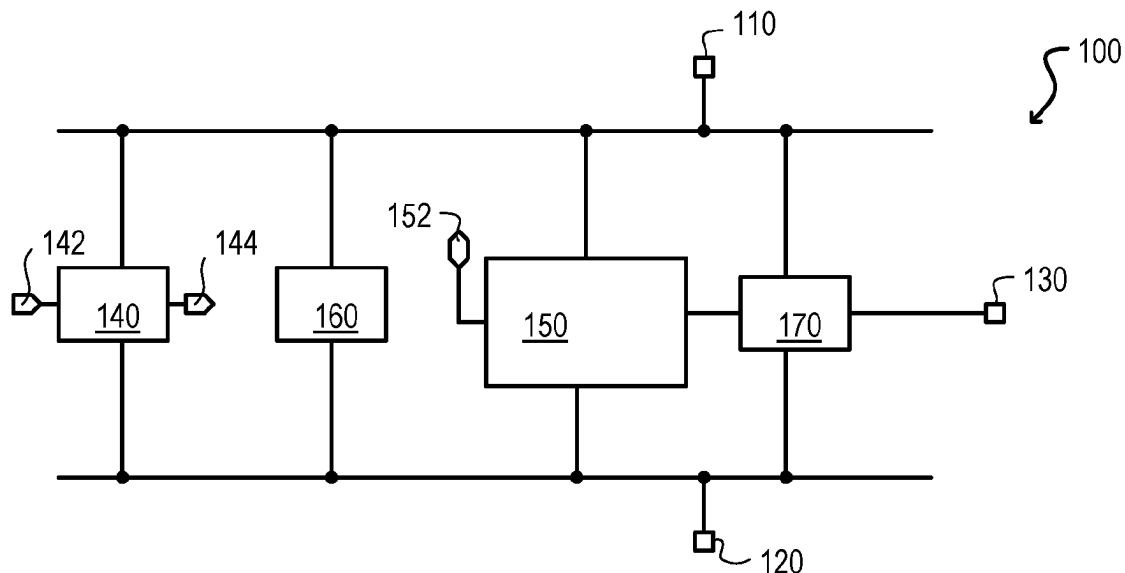
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(57) **ABSTRACT**

An integrated circuit device having insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure has been disclosed. The integrated circuit device may include electrostatic discharge (ESD) protection circuit structures. The ESD protection circuit structures may be formed in regions other than the region that the IGFETs are formed as well as in the region that the IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure are formed. By forming ESD protection circuit structures in regions below the IGFETs, an older process technology may be used and device size may be decreased. Furthermore, planar IGFETs of FinFETs may be formed in other regions to decrease device size and improve costs.

20 Claims, 12 Drawing Sheets



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(2025.01); *H10D 89/921* (2025.01); *H10D*
89/931 (2025.01)

(56) **References Cited**

U.S. PATENT DOCUMENTS

9,219,005	B2	12/2015	Or-Bach et al.
9,711,500	B1	7/2017	Walker
9,871,034	B1	1/2018	Or-Bach
10,290,626	B1	5/2019	Li
10,354,995	B2	7/2019	Or-Bach et al.
11,145,657	B1	10/2021	Or-Bach
11,368,016	B2	6/2022	Walker
11,594,618	B2	2/2023	Chen
2004/0256675	A1	12/2004	McNeil
2005/0247979	A1	11/2005	Fung
2005/0286188	A1 *	12/2005	Camp H01L 27/0262 361/56
2006/0082940	A1 *	4/2006	Lin H02H 9/046 361/91.1
2009/0278251	A1	11/2009	Tsai et al.
2009/0289304	A1	11/2009	Pouydebasque
2012/0162836	A1	6/2012	Furuta et al.
2013/0294170	A1	11/2013	Sivero
2016/0372920	A1	12/2016	Kinzer
2017/0323882	A1 *	11/2017	Langguth H01L 27/1207
2018/0138289	A1	5/2018	Rachmady
2018/0212023	A1	7/2018	Weber

2018/0219007	A1	8/2018	Shrivastava
2018/0247929	A1	8/2018	Shrivastava
2019/0074297	A1	3/2019	Kishishita
2019/0165186	A1	5/2019	Shimbo
2019/0229011	A1 *	7/2019	Suk H01L 29/66439
2020/0168715	A1	5/2020	Wu
2020/0295127	A1	9/2020	Mannebach
2020/0312960	A1	10/2020	Bomberger
2020/0343387	A1	10/2020	Liaw
2020/0403007	A1	12/2020	Thomson
2021/0225839	A1	7/2021	Lin
2021/0296306	A1	9/2021	Walker
2021/0296307	A1	9/2021	Walker
2021/0296889	A1	9/2021	Walker
2021/0305436	A1	9/2021	Kar
2021/0366846	A1	11/2021	Hung
2021/0384188	A1	12/2021	Kim
2021/0391320	A1	12/2021	Wang
2021/0408285	A1	12/2021	Hickey
2022/0044104	A1	2/2022	Liu
2022/0328411	A1	10/2022	Or-Bach

OTHER PUBLICATIONS

U.S. Appl. No. 17/030,662, Advisory Action, dated Aug. 24, 2023.
U.S. Appl. No. 18/125,465, Non-Final Office Action, dated Aug. 1, 2023.
U.S. Appl. No. 18/125,465, Notice of Allowance, dated Aug. 28, 2023.
Office Action, dated Mar. 3, 2023, for U.S. Appl. No. 17/030,662, filed Sep. 24, 2023.
U.S. Appl. No. 62/991,157, filed Mar. 18, 2020, to which the present application claims priority.
U.S. Appl. No. 17/030,694, filed Sep. 24, 2020, to which the present application claims priority.

* cited by examiner

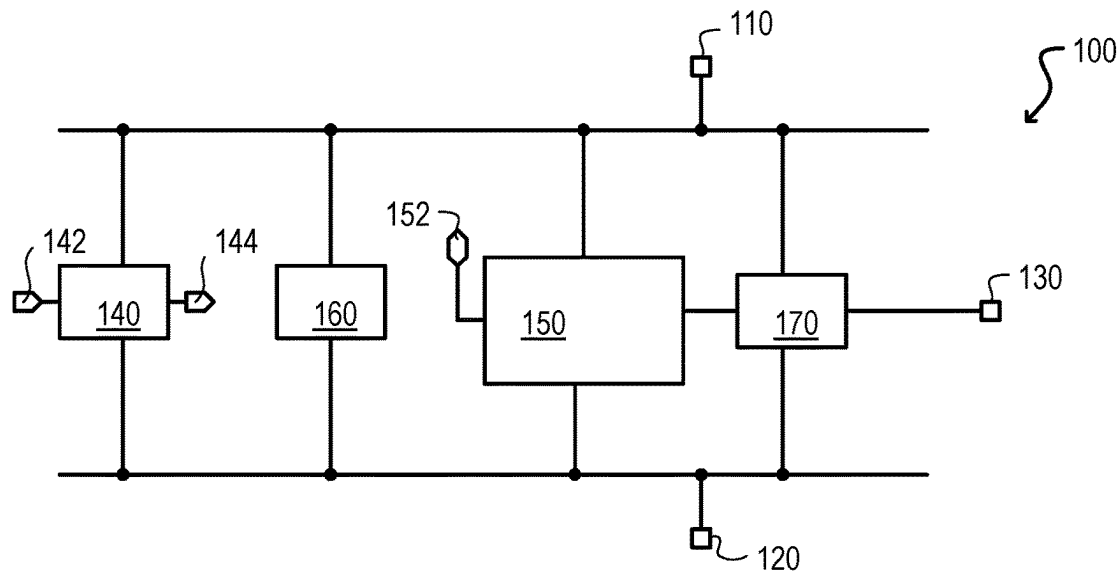


FIG. 1

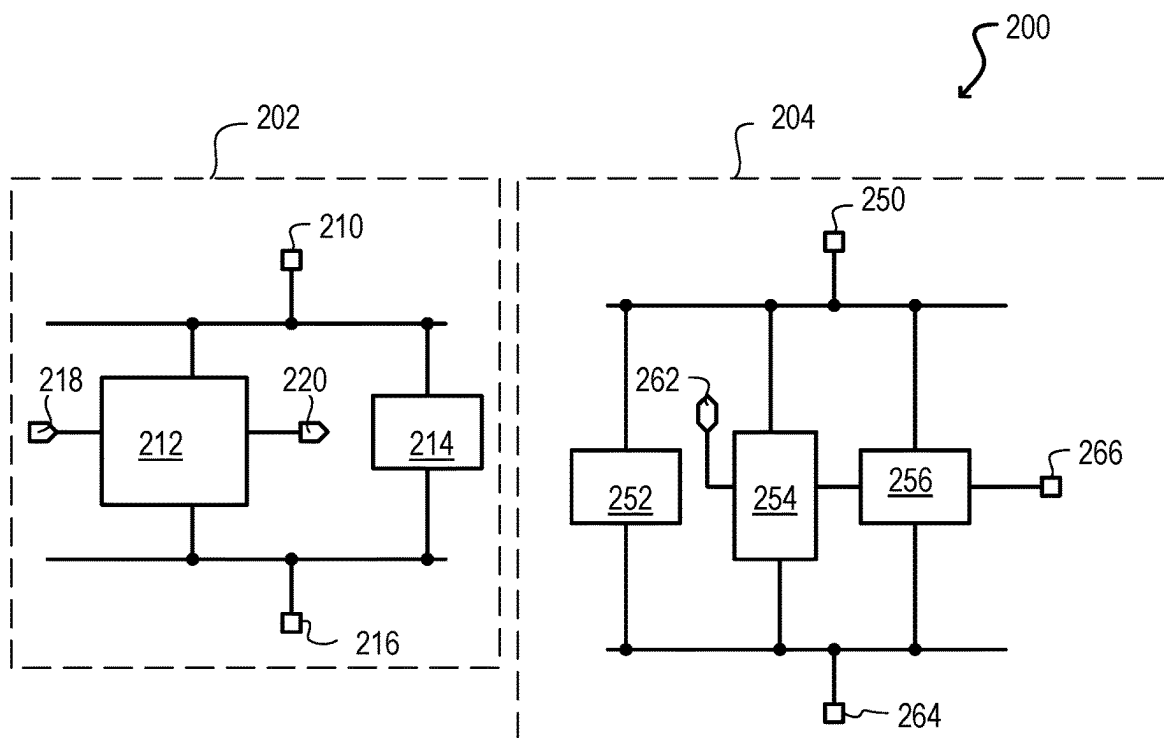


FIG. 2

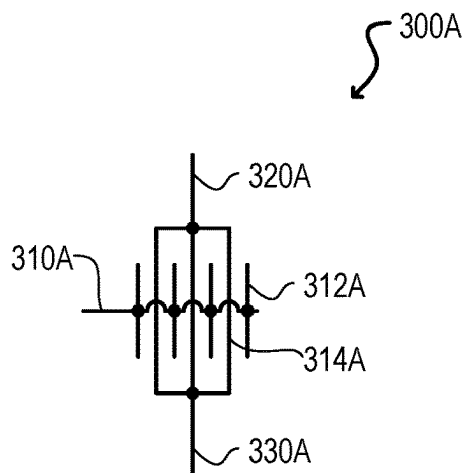


FIG. 3A

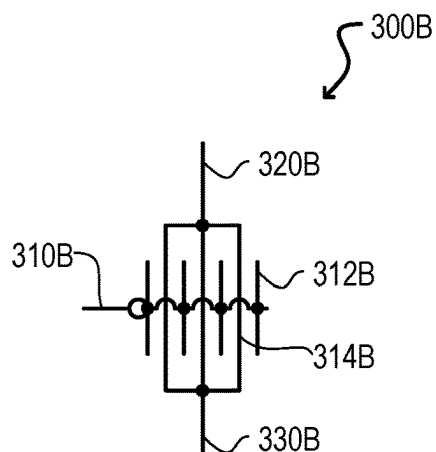


FIG. 3B

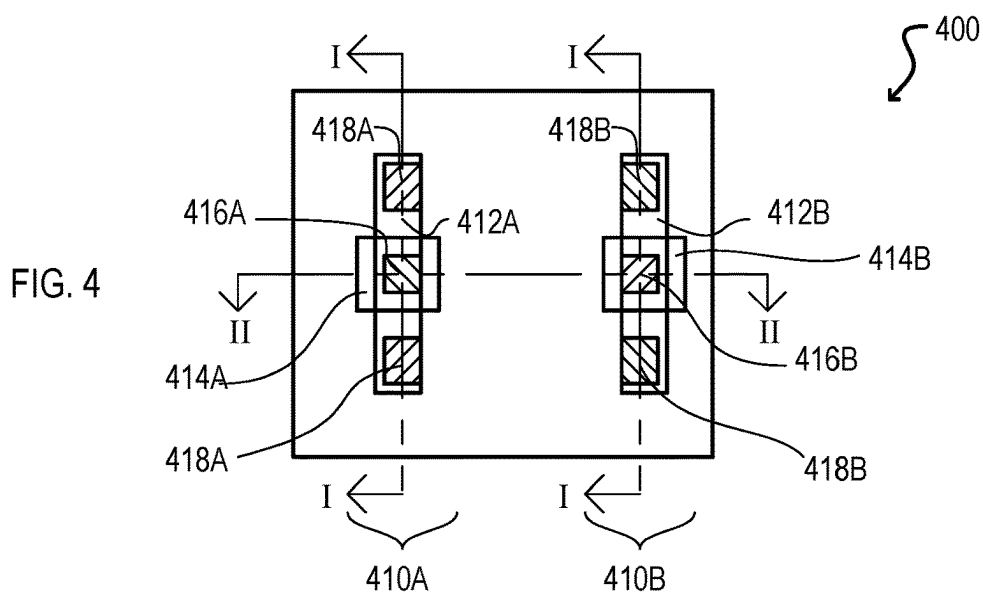


FIG. 4

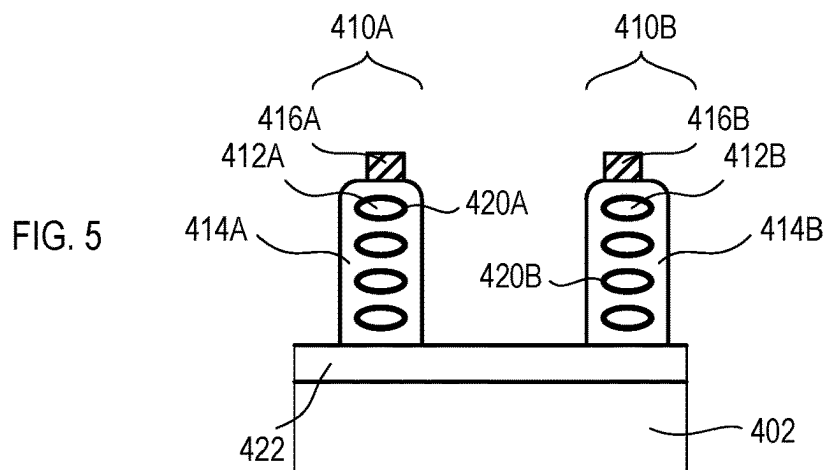
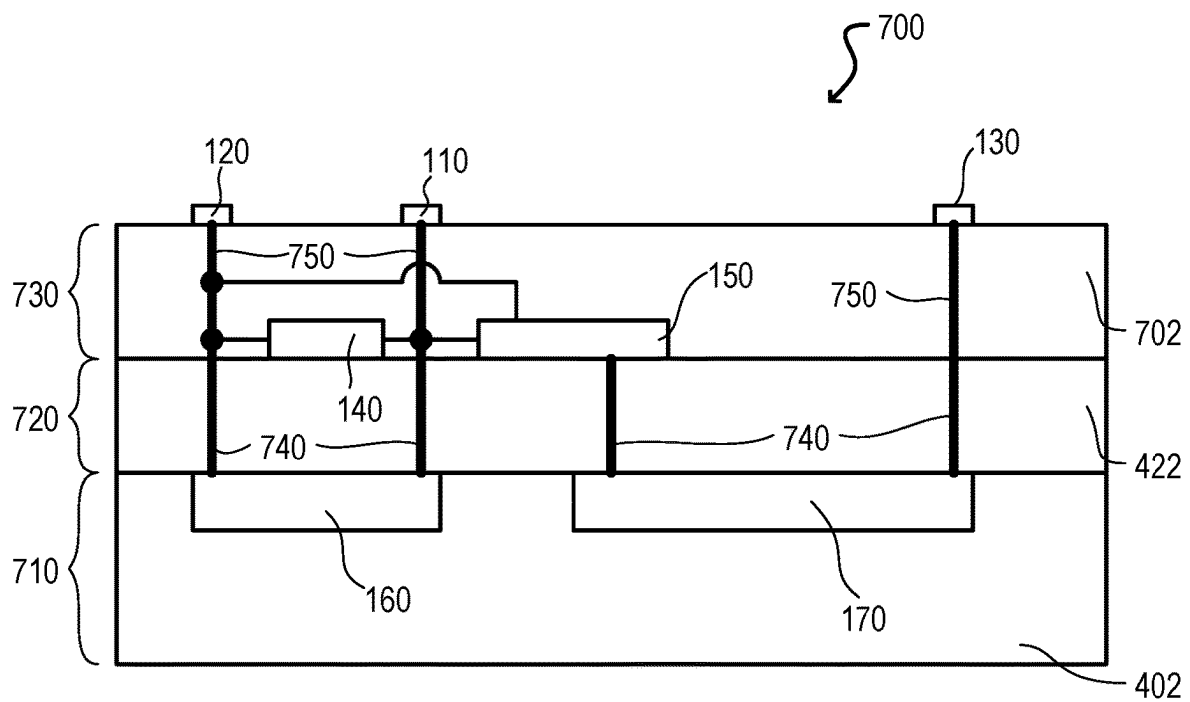
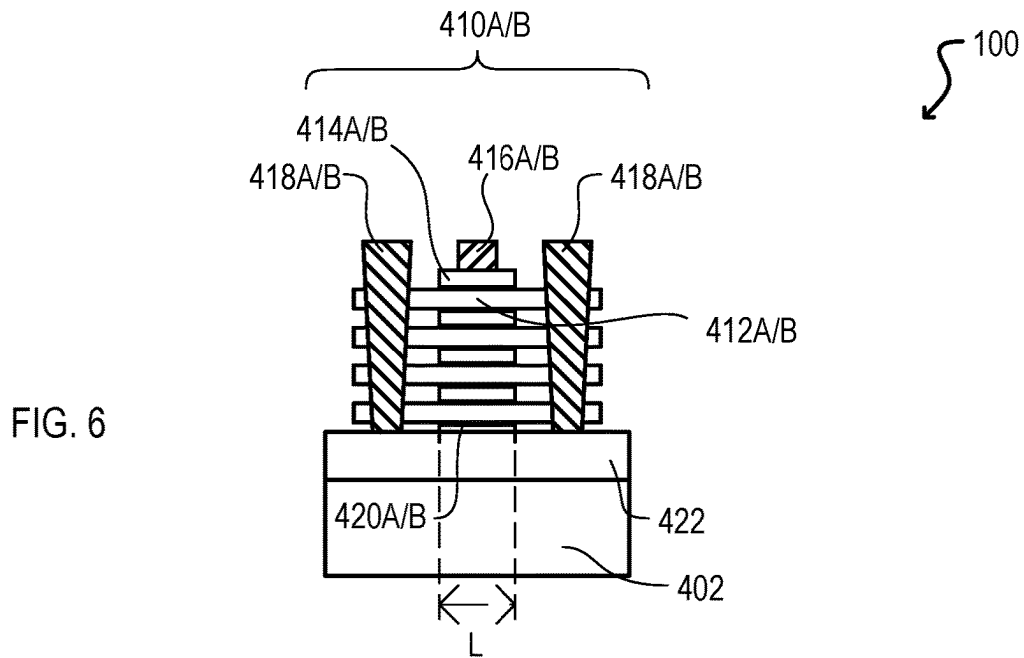


FIG. 5



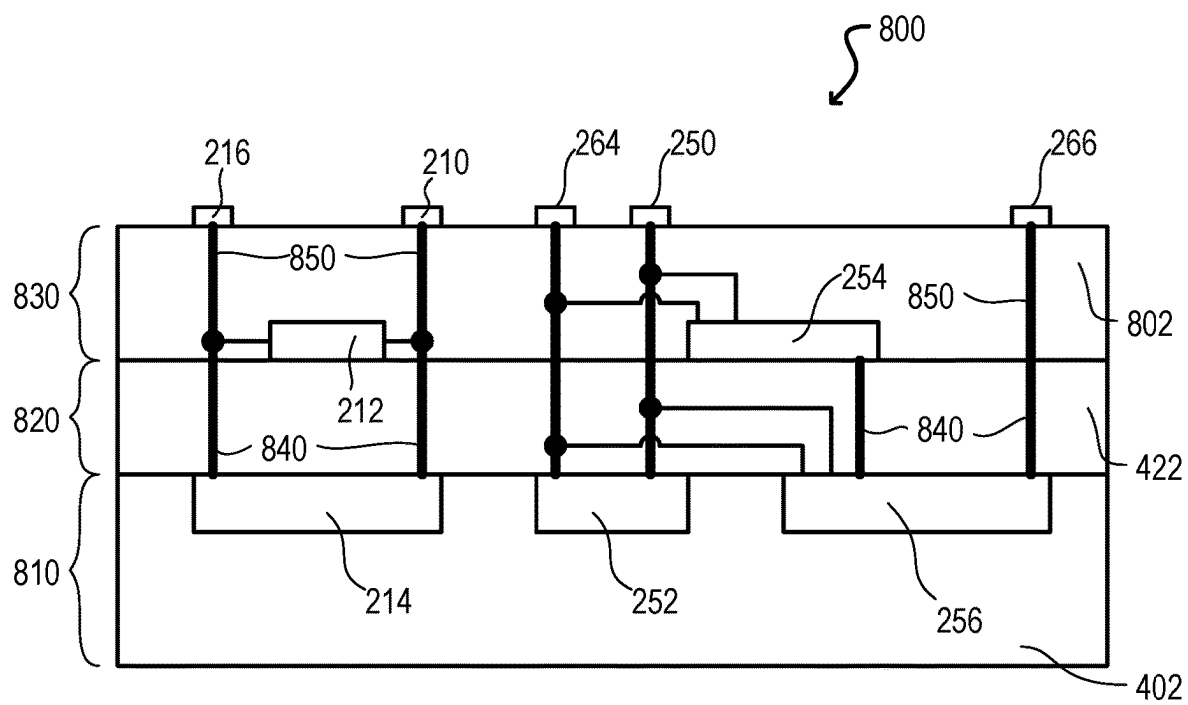


FIG. 8

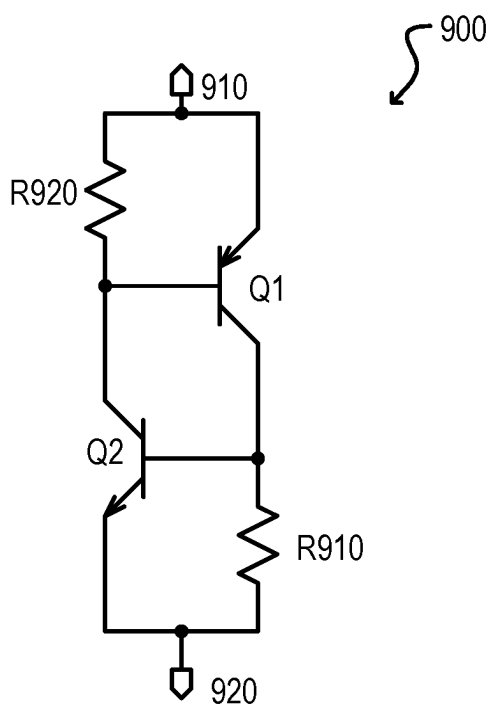


FIG. 9

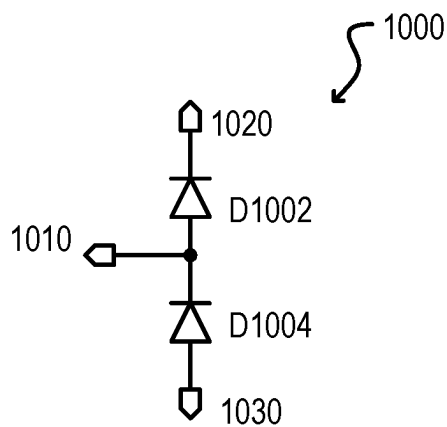


FIG. 10

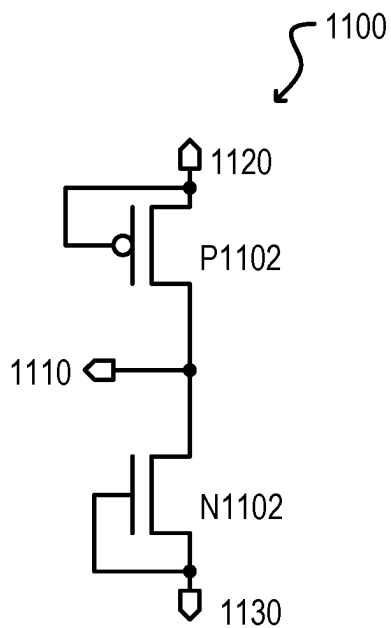


FIG. 11

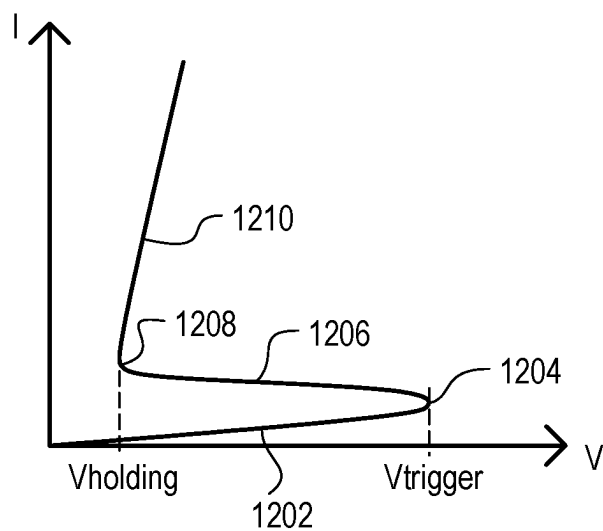
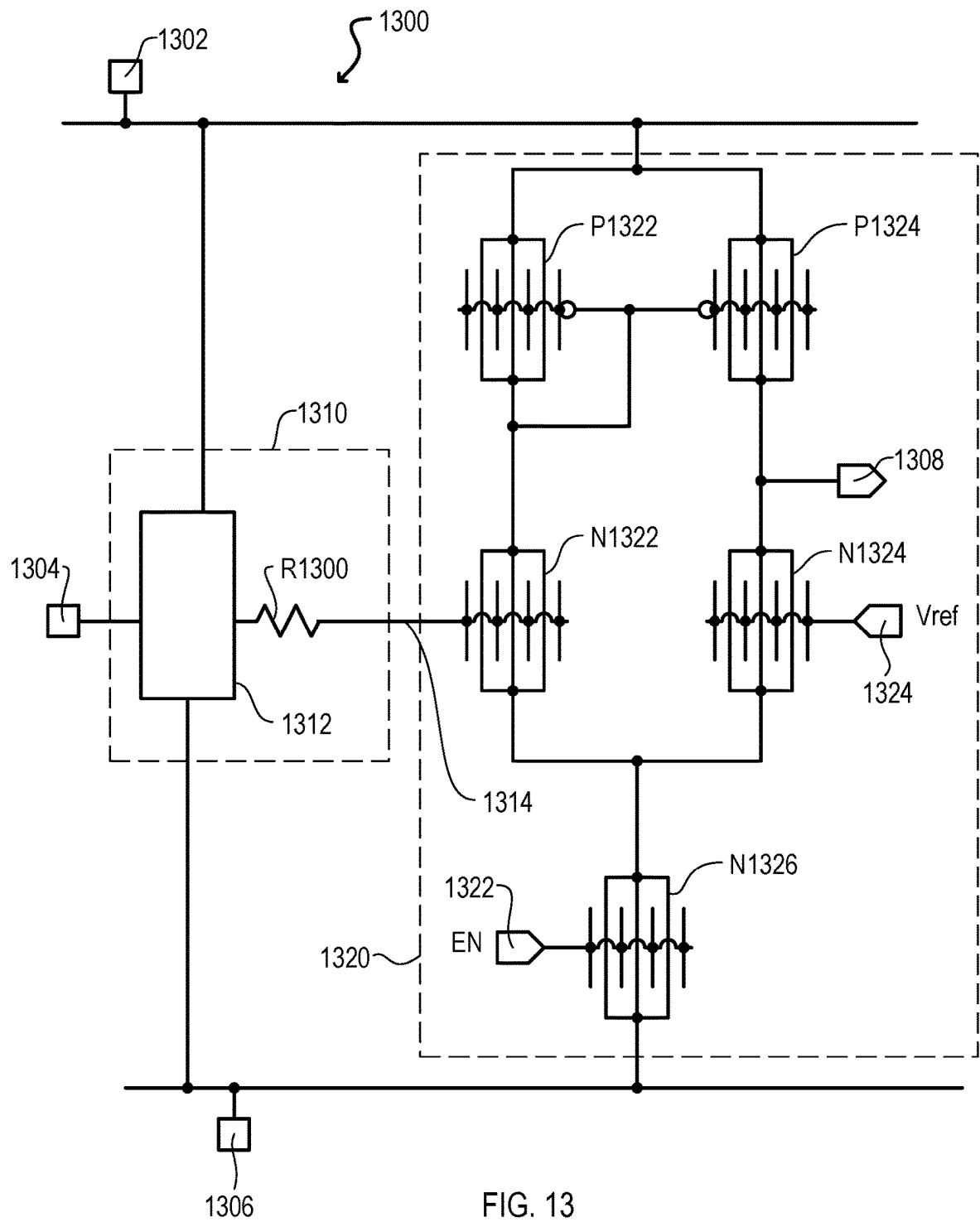
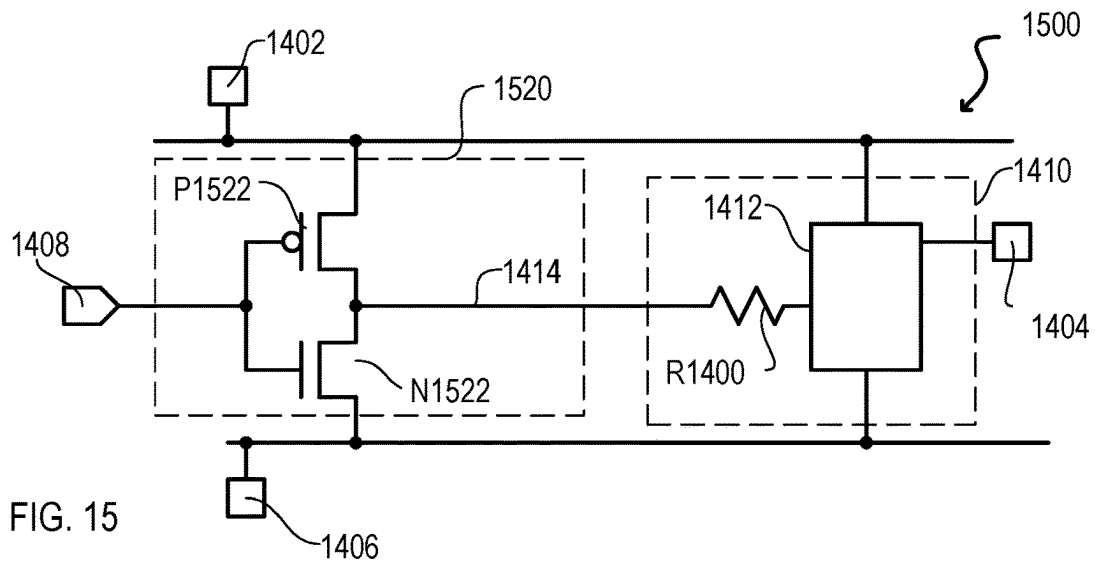
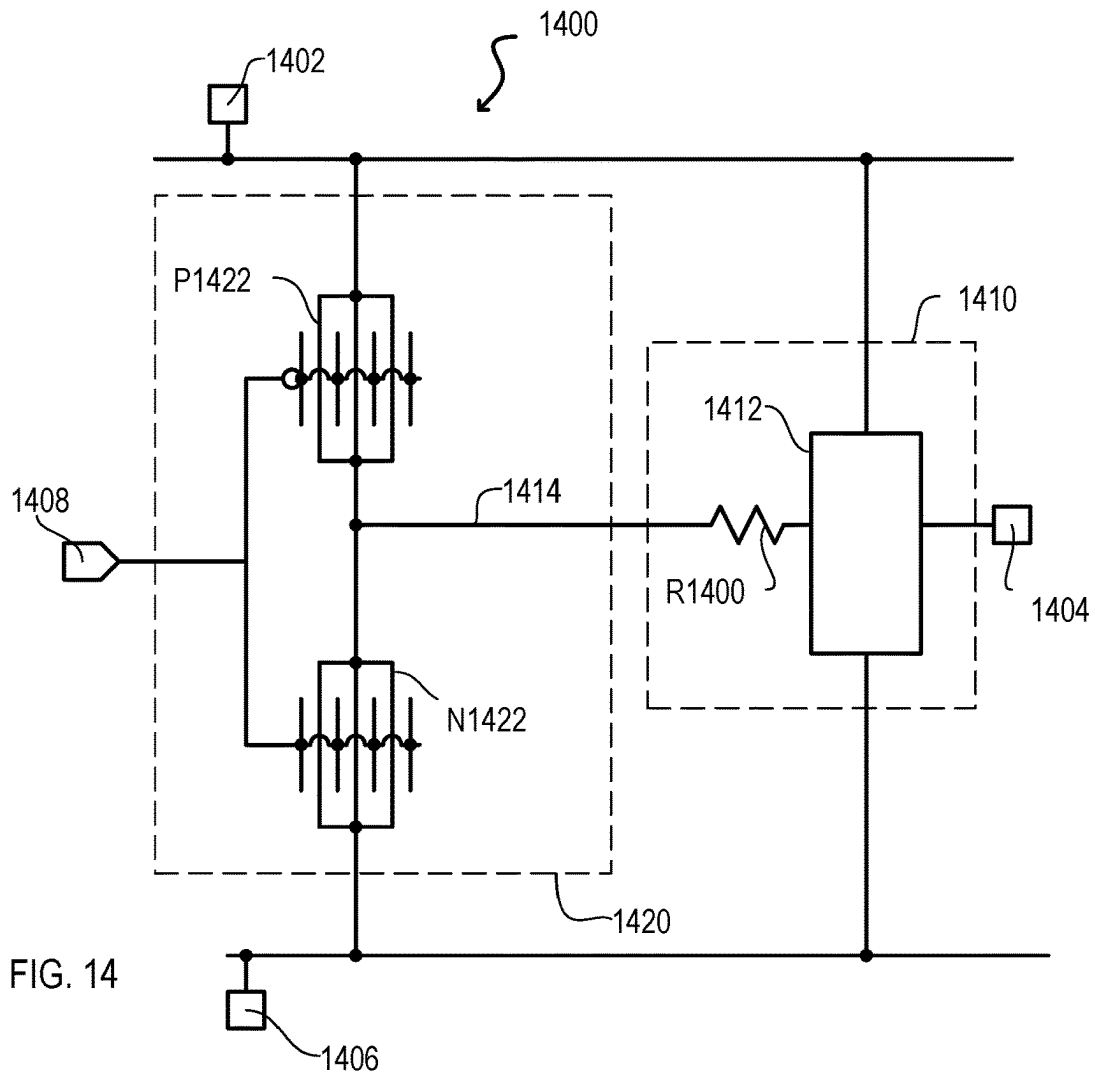


FIG. 12





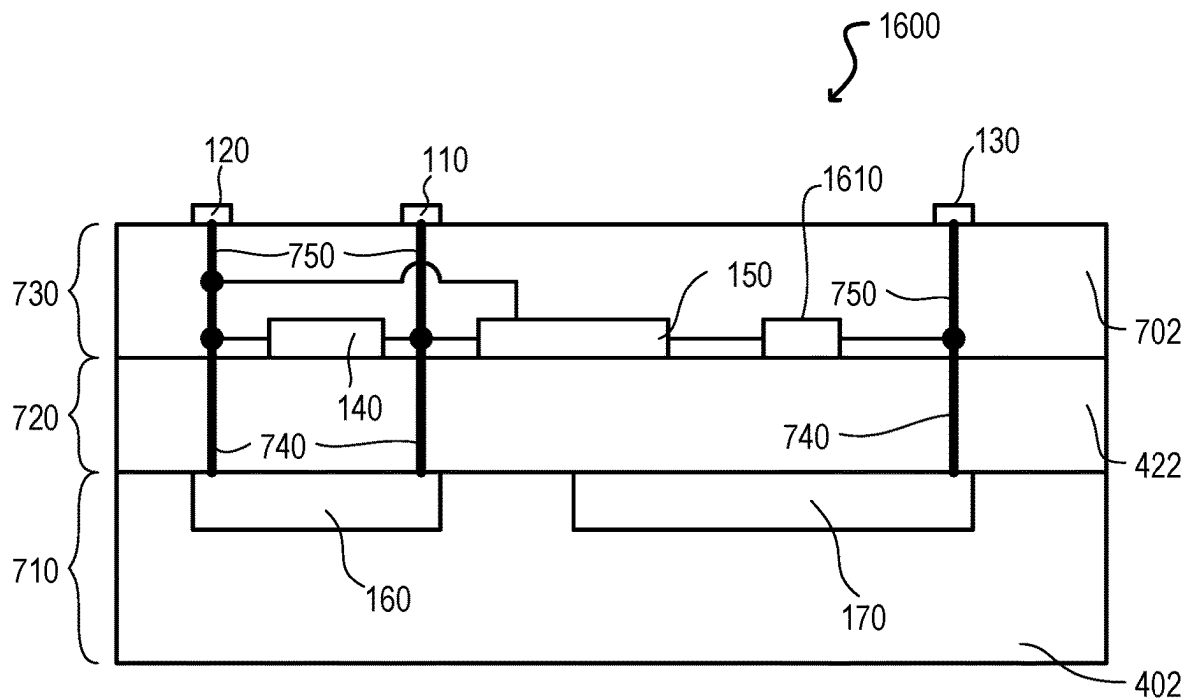


FIG. 16

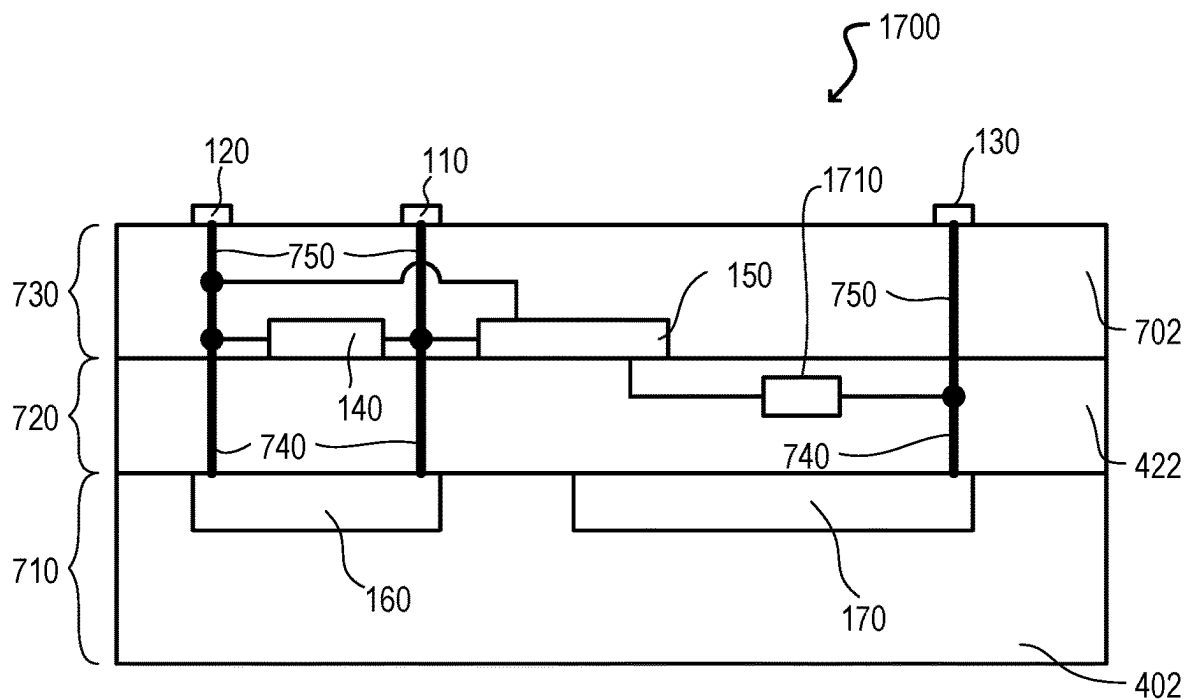


FIG. 17

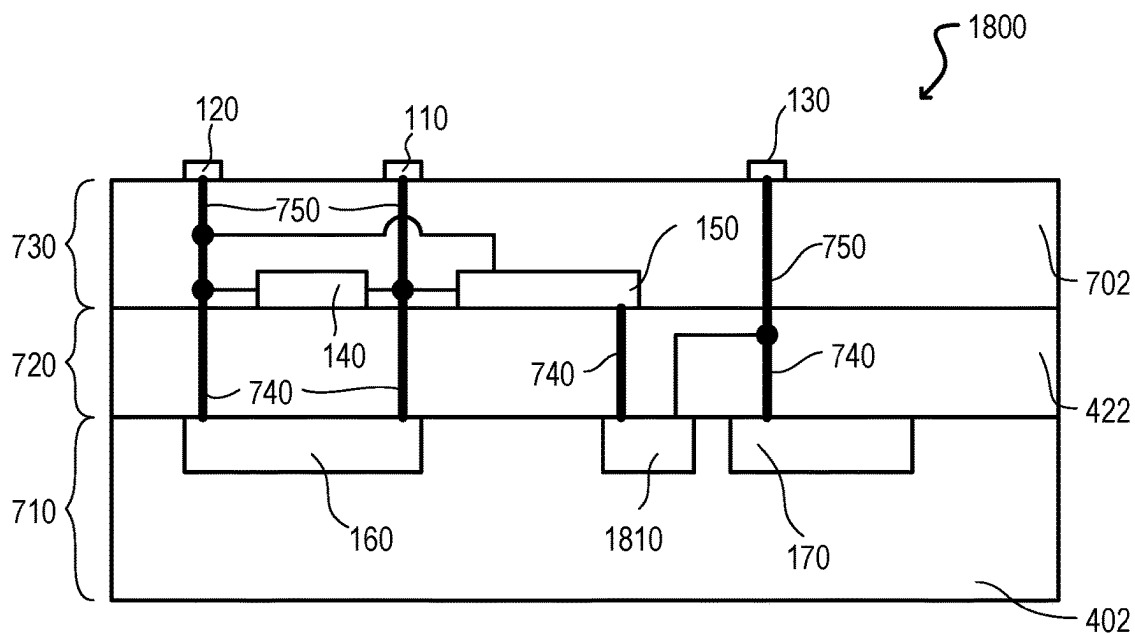


FIG. 18

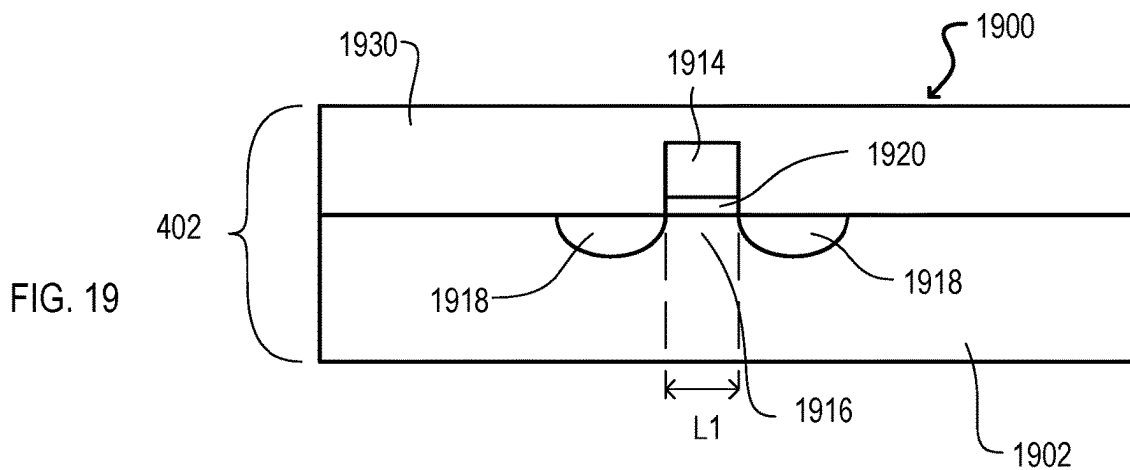


FIG. 19

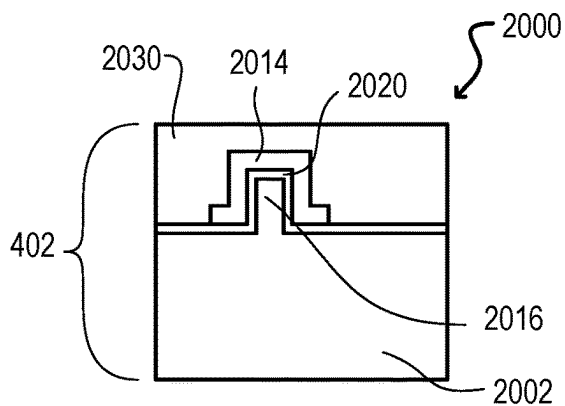


FIG. 20A

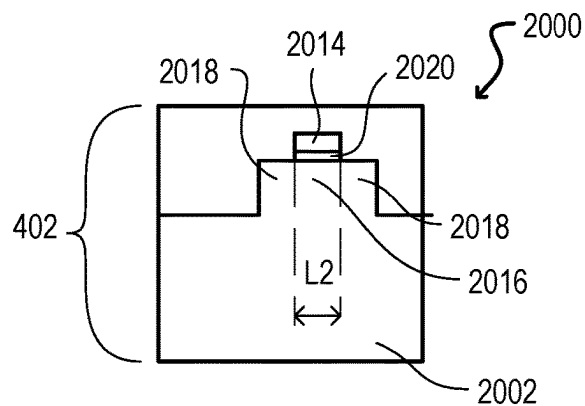


FIG. 20B

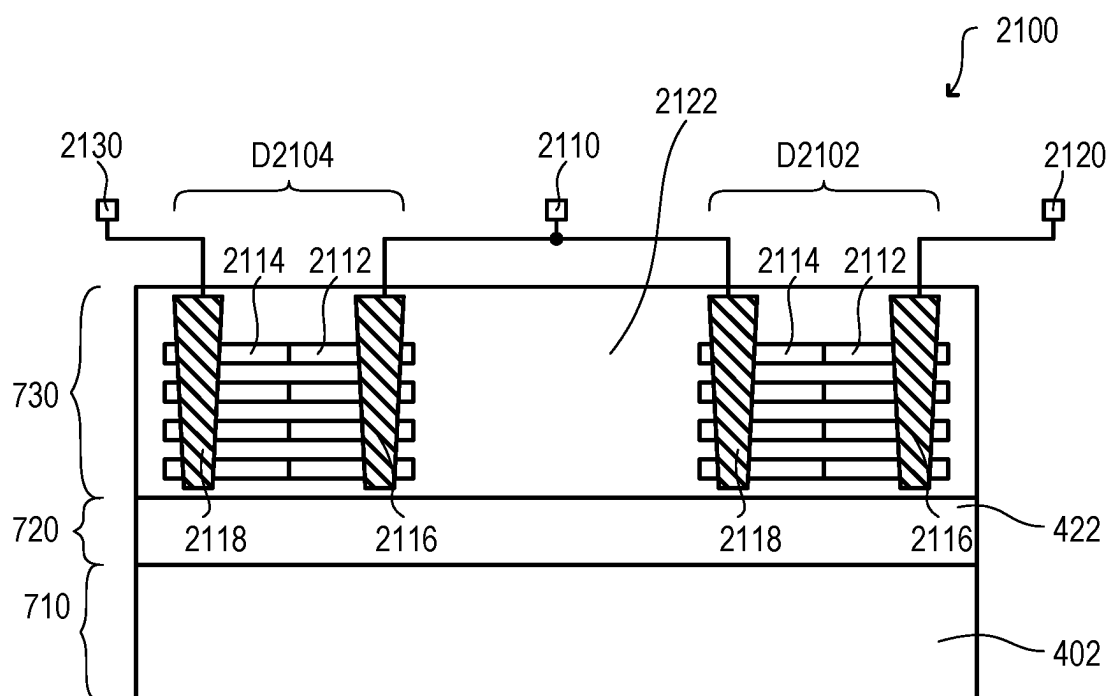


FIG. 21

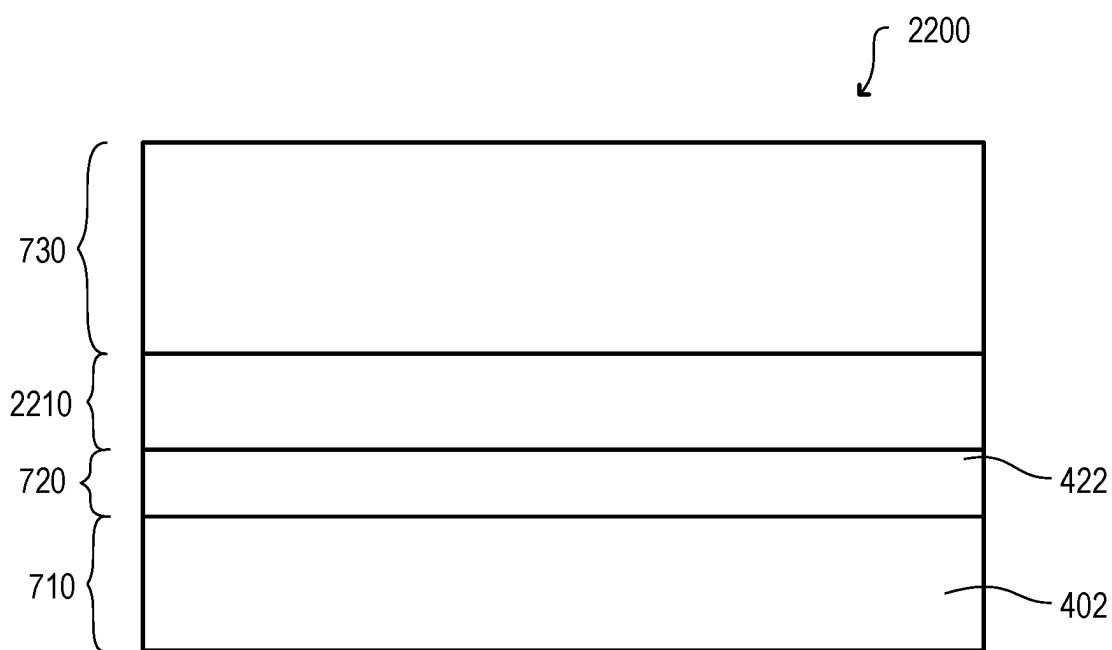


FIG. 22

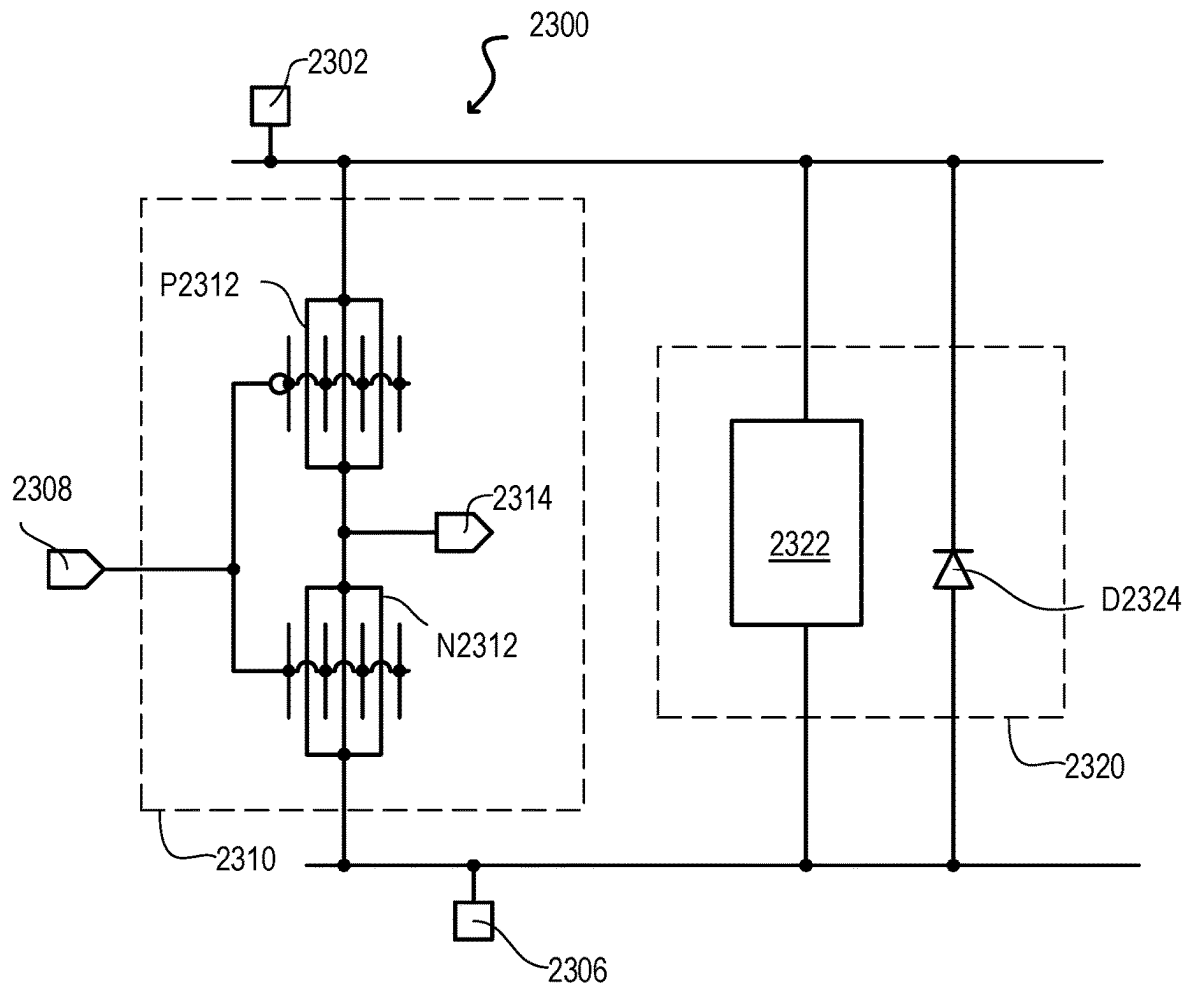


FIG. 23

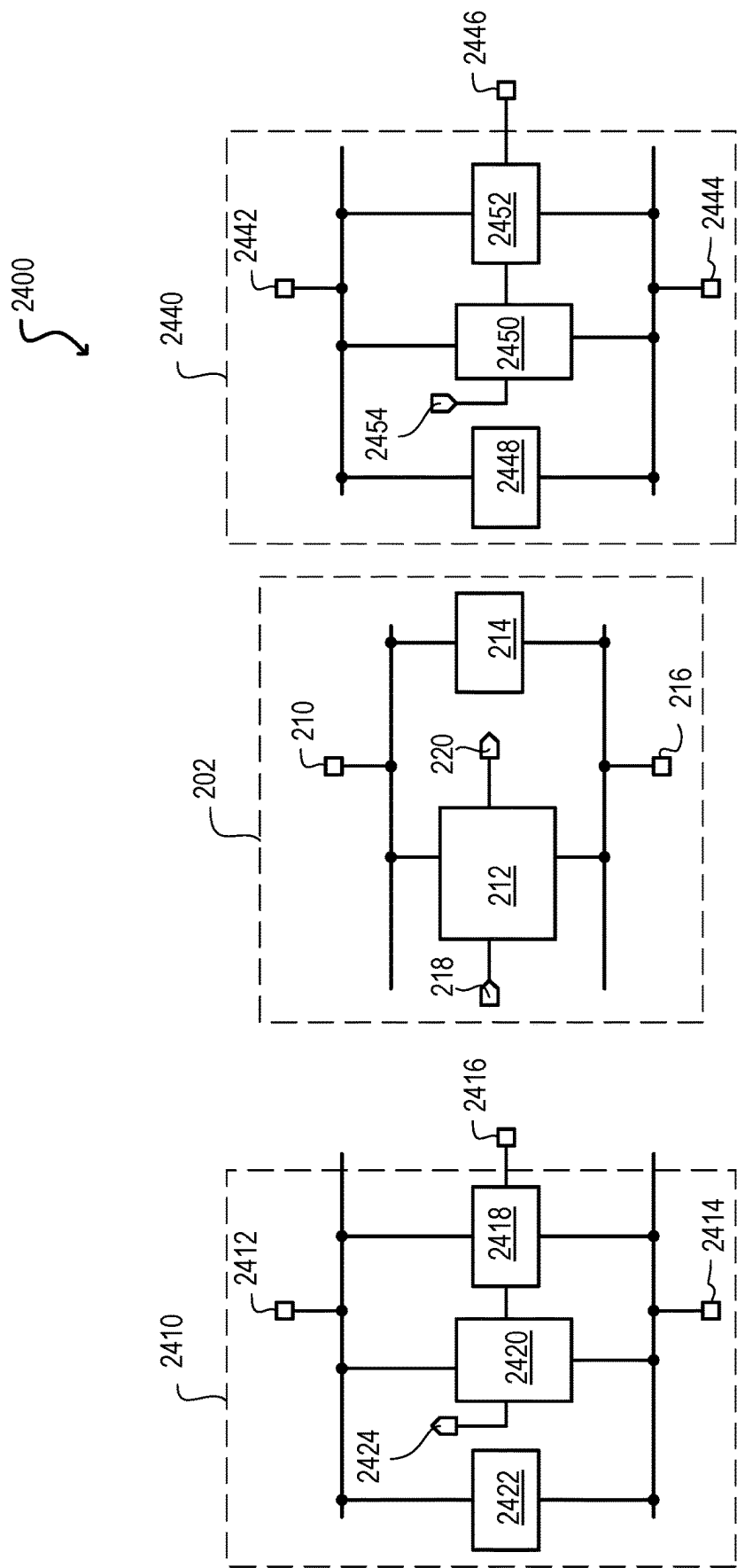


FIG. 24

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ESD PROTECTION FOR INTEGRATED CIRCUIT DEVICES

This application is a continuation of U.S. patent application Ser. No. 17/030,694 filed on Sep. 24, 2020, which claims the benefit of U.S. Provisional Patent Application Ser. No. 62/991,157, filed Mar. 18, 2020, the contents all of which are incorporated by reference herein.

TECHNICAL FIELD

The present invention relates generally to an integrated circuit (IC) device, and more particularly to improving electrostatic discharge (ESD) protection for an IC device.

BACKGROUND OF THE INVENTION

As transistor sizes get smaller, electrostatic discharge (ESD) can be more problematic due to smaller gate dielectric thicknesses and shorter transistor channels. Furthermore, ESD protection circuit structures may be incompatible with new technology and/or consume too much of the active footprint of an integrated circuit device.

In light of the above, it would be desirable to provide ESD protection circuit structures having current discharge capabilities being integrated with new device technology and having a smaller footprint effect on an integrated circuit device.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a circuit schematic diagram of an integrated circuit device including a circuit having an ESD protection circuit structure according to an embodiment.

FIG. 2 is a circuit schematic diagram of an integrated circuit device including a circuit having an ESD protection circuit structure according to an embodiment.

FIGS. 3A and 3B are circuit schematic diagrams of complementary IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure according to an embodiment.

FIG. 4 is a top plan view of an integrated circuit device including n-type and p-type IGFETs according to an embodiment.

FIG. 5 is a cross sectional view of integrated circuit device according to an embodiment.

FIG. 6 is a cross sectional view of integrated circuit device according to an embodiment.

FIG. 7 is a schematic diagram of an integrated circuit device according to an embodiment.

FIG. 8 is a schematic diagram of an integrated circuit device according to an embodiment.

FIG. 9 is a circuit schematic diagram of an ESD protection circuit structure according to an embodiment.

FIG. 10 is a circuit schematic diagram of an ESD protection circuit structure according to an embodiment.

FIG. 11 is a circuit schematic diagram of an ESD protection circuit structure according to an embodiment.

FIG. 12 is a current-voltage diagram of an ESD protection circuit structure according to an embodiment.

FIG. 13 is a circuit schematic diagram of an integrated circuit device according to an embodiment.

FIG. 14 is a circuit schematic diagram of an integrated circuit device according to an embodiment.

FIG. 15 is a circuit schematic diagram of an integrated circuit device according to an embodiment.

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FIG. 16 is a circuit schematic diagram of an integrated circuit device according to an embodiment.

FIG. 17 is a schematic diagram of an integrated circuit device according to an embodiment.

FIG. 18 is a schematic diagram of an integrated circuit device according to an embodiment.

FIG. 19 is a cross-sectional schematic diagram of a planar IGFET that can be formed in a region according to an embodiment.

FIGS. 20A and 20B are cross-sectional schematic diagrams of a FinFET that can be formed in a region according to an embodiment.

FIG. 21 is a schematic diagram of an integrated circuit device having an ESD protection circuit structure having a plurality of horizontally current carrying regions that can be vertically aligned above a substrate according to an embodiment.

FIG. 22 is a diagram of an integrated circuit device according to an embodiment.

FIG. 23 is a circuit schematic diagram of an internal circuit and an ESD protection circuit structure according to an embodiment.

FIG. 24 is a circuit schematic diagram of an integrated circuit device including a circuit having an ESD protection circuit structure according to an embodiment.

DETAILED DESCRIPTION OF THE EMBODIMENTS

According to the embodiments set forth below, an input circuit, output circuit, and or input and output circuit including a transistor having a plurality of vertically stacked channels with improved gate control can be electrically connected to an external terminal and have an ESD (electrostatic discharge) circuit having at least portion of the ESD protection circuit structure formed in a layer/region below the input/output circuit. In this way, the footprint of the ESD structure as well as current leakage may be reduced and reliable ESD may be maintained.

Referring now to FIG. 1, an integrated circuit device including a circuit having an ESD protection circuit structure according to an embodiment is set forth in a schematic diagram and given the general reference character 100.

The circuit formed on integrated circuit device 100 can include pads (110, 120, and 130), internal circuit 140, interface circuit 150, and ESD protection circuit structures (160 and 170). Integrated circuit device 100 may be a semiconductor device.

Pad 110 may receive an externally provided supply potential (for example VDD). Pad 120 may receive an externally provided power supply potential (for example, VSS i.e. ground potential).

Pad 110 may be electrically connected to provide an externally provided supply potential (for example VDD) to internal circuit 140, interface circuit 150, and ESD protection circuit structures (160 and 170). Pad 120 may be electrically connected to provide an externally provided power supply potential (for example VSS) to internal circuit 140, interface circuit 150, and ESD structures (160 and 170). Pad 130 may provide and/or receive an external signal (for example, a data or control signal) to or from interface circuit 150 through ESD protection circuit structure 170. Interface circuit may receive or generate an internal signal at terminal 152. Interface circuit 150 may be a buffer circuit that buffers the external and internal signals generated or received.

Internal circuit 140 may receive at least one signal at an input terminal 142 and may provide at least one signal at an

output terminal **144**. In another example, internal circuit **140** may be an internal voltage regulating circuit that provides an internal power supply potential. The input terminal **142** and the output terminal **144** of internal circuit **140** are not electrically connected to any pad that can receive or provide a signal external to the integrated circuit device **100**.

In one embodiment, internal circuit **140** may include insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure. Interface circuit **150** may include insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure. Both internal circuit **140** and interface circuit **150** may include p-type and n-type IGFETs. In one embodiment, ESD protection circuit structures (**160** and **170**) may include electrical components (such as diodes) formed with a plurality of horizontally disposed cathodes and anodes that can be vertically aligned above a substrate. In one embodiment, ESD protection circuit structures (**160** and **170**) may include electrical components (such as diodes, transistors, silicon controlled rectifiers (SCRs) and/or resistors) formed in the substrate. In one embodiment interface circuit **150** may include electrical components (such as IGFETs) formed in the substrate.

When ESD protection circuit structures (**160** and **170**) are formed in a semiconductor substrate of integrated circuit device **100**, a process having larger critical dimensions (i.e. an older process and less expensive) may be used. The semiconductor substrate may then be sent to a state of the art fabrication facility to form the circuit including insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure as will be discussed further in the specification.

Referring now to FIG. **2**, an integrated circuit device including a circuit having an ESD protection circuit structure according to an embodiment is set forth in a circuit schematic diagram and given the general reference character **200**. Integrated circuit device **200** may include a first circuit section **202** and a second circuit section **204**. First circuit section **202** may include circuits that only have external connections to a power supply potential and/or a ground (VSS) potential. Second circuit section **204** may include circuits that have external connections to a power supply potential, a ground potential, and/or a pad coupled to provide or receive an external signal, such as a data signal, control signal or a clock signal, as just a few examples.

First circuit section **202** may include an internal circuit **212** and an ESD protection circuit structure **214**. Internal circuit **212** and ESD structure **214** may each be electrically connected to pad (**210** and **216**). Internal circuit **212** may receive an input signal at an input terminal **218** and may provide an output signal at an output terminal **220**. Pad **210** may receive an external power supply potential, such as VDD and pad **216** may receive an external reference potential such as VSS. In other embodiments, internal circuit **212** may be an internal power supply generator and may receive an external power supply potential at pad **210** and may provide an internal power supply potential to be used by internal circuits. The input terminal **218** and the output terminal **220** of internal circuit **212** are not electrically connected to any pad that can receive or provide a signal external to the integrated circuit device **200**.

Second circuit section **204** may include pads (**250**, **264**, and **266**), an ESD protection circuit structure **252**, an interface circuit **254**, and ESD protection circuit structure **256**. Interface circuit **254** may provide or receive an internal signal at terminal **262** and may provide and/or receive an external signal at pad **266** through ESD protection circuit structure **256**. Interface circuit **254** may be electrically connected to pads (**250** and **264**). Pads (**250** and **264**) may respectively receive an external power supply potential (such as VDD) and a reference potential (such as VSS). ESD structure **252** may be electrically connected between pads (**250** and **264**). ESD structure **256** may be electrically connected to pads (**250**, **264**, and **266**).

In one embodiment, internal circuit **212** may include insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure. Interface circuit **254** may include insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure. Both internal circuit **212** and interface circuit **254** may include p-type and n-type IGFETs. In one embodiment, ESD protection circuit structures (**214**, **252**, and **256**) may include electrical components (such as diodes, transistors, and/or resistors) formed with a plurality of horizontally disposed cathodes and anodes that can be vertically aligned above a substrate. In one embodiment, ESD protection circuit structures (**214**, **252**, and **256**) may include electrical components (such as diodes, transistors, SCRs and/or resistors) formed in the substrate. In one embodiment interface circuit **254** may include electrical components (such as IGFETs) formed in the substrate.

When ESD protection circuit structures (**214**, **252**, and **256**) are formed in a semiconductor substrate of integrated circuit device **200**, a process having larger critical dimensions (i.e. an older and cheaper process) may be used. The semiconductor substrate may then be sent to a state of the art fabrication facility to form the circuit including insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure as will be discussed further in the instant specification.

Power supply potentials externally provided to pads (**210** and **250**) may be different power supply potentials, such as a first potential (VDD1) for internal circuit **212** and a second potential (VDD2) for interface circuit **254**.

The IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure will now be discussed.

Referring now to FIGS. **3A** and **3B**, circuit schematic diagrams of complementary IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure according to an embodiment are set forth. FIG. **3A** is a N-channel (N-type) IGFET **300A** and FIG. **3B** is a P-channel (P-type) IGFET **300B**.

N-channel IGFET **300A** includes a control gate terminal **310A**, a first source/drain terminal **320A**, and a second source/drain terminal **330A**. Control gate terminal **310A** may be electrically connected to control gate **312A**. Control gate **312A** may be drawn as a plurality of control gates on each side of a plurality of channel region **314A**. In reality,

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control gate **312A** may surround a plurality of horizontally disposed channel regions **314A** that can be vertically aligned above a substrate.

P-channel IGFET **300B** includes a control gate terminal **310B**, a first source/drain terminal **320B**, and a second source/drain terminal **330B**. Control gate terminal **310B** may be electrically connected to control gate **312B**. Control gate **312B** may be drawn as a plurality of control gates on each side of a plurality of channel region **314B**. In reality, control gate **312B** may surround a plurality of horizontally disposed channel regions **314B** that can be vertically aligned above a substrate.

Referring now to FIG. 4, a top plan view of an integrated circuit device including n-type and p-type IGFETS according to an embodiment is set forth and given the general reference character **400**.

Integrated circuit device **400** may include an N-type IGFET **410A** and a P-type IGFET **410B**.

N-type IGFET **410A** and P-type IGFET **410B** may each include a control gate that may surround a plurality of horizontally disposed channel regions that can be vertically aligned above a substrate.

N-type IGFET **410A** may include drain/source contacts **418A**, a gate contact **416A**, a gate structure **414A**, and vertically aligned and horizontally disposed channel region structures **412A**.

P-type IGFET **410B** may include drain/source contacts **418B**, a gate contact **416B**, a gate structure **414B**, and vertically aligned and horizontally disposed channel region structures **412B**.

Referring now to FIG. 5, a cross sectional view of integrated circuit device **400** is set forth. The cross-sectional view is along the line II-II of FIG. 4.

Integrated circuit device **400** may include a substrate **402**, an insulator layer **422** a N-type IGFET **410A**, and a P-type IGFET **410B**.

N-type IGFET **410A** may include a gate contact **416A**, a gate structure **414A**, and vertically aligned and horizontally disposed channel regions **412A**, and gate insulating layer **420A**. Gate insulating layer **420A** may surround each vertically aligned and horizontally disposed channel regions **412A**.

P-type IGFET **410B** may include a gate contact **416B**, a gate structure **414B**, and vertically aligned and horizontally disposed channel regions **412B**, and gate insulating layer **420B**. Gate insulating layer **420B** may surround each vertically aligned and horizontally disposed channel regions **412B**.

As will be discussed later, IGFETS including vertically aligned and horizontally disposed channel region structures may be used in internal circuits **140** of FIG. 1 or internal circuits **212** of FIG. 2 and/or interface circuit **150** of FIG. 1 and/or interface circuit **254** of FIG. 2, for example and ESD protection circuit structures (**140**, **160**, **170**, **214**, **252**, and/or **256**) may include diodes, transistors, SCRs and/or resistors formed in substrate **402**.

Referring now to FIG. 6, a cross sectional view of integrated device **400** is set forth. The cross-sectional view is along the line I-I of FIG. 4. As shown in FIG. 4, there are two lines I-I as the N-type IGFET **410A** and P-type IGFET **410B** may have similar structures except the materials and/or doping of materials may differ and elements are designated with the suffix "A/B" to illustrate such. Semiconductor device **400** may include a substrate **402**, an insulator layer **422** and N-type and P-type IGFETS (**410A/B**). IGFET **410A/B** may include a gate contact **416A/B**, a gate structure **414A/B**, vertically aligned and horizontally

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disposed channel regions **412A/B**, gate insulating layer **420A/B**, and drain/source contacts **418A/B**. Gate structure **416A/B** and gate insulating layer **420A/B** may surround each vertically aligned and horizontally disposed channel regions **412A/B**.

IGFETs (**410A** and **410B**) may be formed by forming a layered crystal of two materials over dielectric region **422**. For example, layers of silicon and silicon germanium may be formed. An etch and deposit step may then be used to form the source/drain regions (**418A** and **418B**). The silicon layer may form the channel regions (**412A** and **412B**). After a vertical etch, the silicon germanium layers may be etched by using a chemical that can selectively etch silicon germanium with the source/drain regions (**418A** and **418B**) used as support structures. Next, the gate dielectric layers (**420A** and **420B**) may be formed using atomic layer deposition, for example of hafnium-dioxide. Then gate structure (**416A** and **416B**) may be formed using atomic layer deposition of a metal layer, for example, tungsten. The n-type IGFETs **410A** may have source/drain regions **418A** doped with n-type carriers, such as phosphorous and/or arsenic, for example. The p-type IGFETs **410B** may have source/drain regions **418B** doped with p-type carriers, such as boron, for example.

As will be discussed later, IGFETS including vertically aligned and horizontally disposed channel region structures may be used in internal circuits **140** of FIG. 1 or internal circuits **212** of FIG. 2 and/or interface circuits **150** of FIG. 1 and/or interface circuit **254** of FIG. 2, for example and ESD structures (**140**, **160**, **170**, **214**, **252**, and/or **256**) may include diodes, transistors, and/or resistors formed in substrate **402**.

Referring now to FIG. 7, a schematic diagram of an integrated circuit device according to an embodiment is set forth and given the general reference character **700**.

Integrated circuit device **700** may include similar constituents as integrated circuit device **100** including IGFETs of integrated circuit device **400**, such constituents may be given the same reference character. Integrated device **700** can include ESD protection circuit structures **160** and **170** formed in a substrate **402**, an internal circuit **140**, and an interface circuit **150**.

Integrated circuit device **700** may include different regions. A region **710** may include ESD structures (**160** and **170**) formed in a semiconductor substrate **402**. Another region **720** may include an insulator region **422** which may contain wirings **740**. Wirings **740** may provide an interconnect between ESD structures (**160** and **170**) and interface circuit **150**, internal circuit **140**, and/or pads (**110**, **120**, and **130**). Wirings **740** may be in the form of vertical vias that are formed through insulator layer **422** and/or region **720**. Another region **730** may include internal circuit **140** and interface circuit **150**, as well as wirings **750**, and pads (**110**, **120**, and **130**).

Pad **110** may receive an externally provided supply potential (for example VDD). Pad **120** may receive an externally provided power supply potential (for example VSS), and pad **130** may provide and/or receive an external signal (for example, a data or control signal).

ESD protection circuit structures (**160** and **170**) in region **710** may be formed using planar IGFETs, n-type and/or p-type diffusion regions, and silicon control rectifiers (SCR), for example. Region **720** may include passive elements, such as polysilicon and/or metal resistors, incorporated in ESD protection circuit structures (**160** and **170**).

Internal circuit **140** and interface circuit **150** in region **730** may include p-type and n-type IGFETs having a plurality of

horizontally disposed channels that can be vertically aligned above substrate 402. Region 730 may generally have circuitry comprising p-type and n-type IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above substrate 402. Region 710 may include planar IGFETs fabricated using older technologies with more relaxed critical dimensions. In this way, reliable ESD protection circuit structures can be made more cheaply. Another advantage is that the region 730 exclusively has the normal operating circuits (i.e. exclusive of ESD protection circuit structures which only operate when there is an ESD event). For example, if integrated circuit device 700 is a microprocessor, the central processing unit (CPU), bus, and memory would all be located in region 730 and manufactured with a cutting-edge state of the art process having smaller critical dimensions. By forming ESD structure (160 and 170) below the functional circuits, chip size can be reduced.

Referring now to FIG. 8, a schematic diagram of an integrated circuit device according to an embodiment is set forth and given the general reference character 800.

Integrated circuit device 800 may include similar constituents as semiconductor device 200 including IGFETs of integrated circuit device 400, such constituents may be given the same reference character. Integrated circuit device 800 can include ESD protection circuit structures (214, 252, and 256) formed in a substrate 402, an internal circuit 212, and an interface circuit 254.

Integrated circuit device 800 may include different regions. A region 810 may include ESD protection circuit structures (214, 252, and 256) formed in a semiconductor substrate 402. Another region 820 may include an insulator region 422 which may contain wirings 840. Wirings 840 may provide an interconnect between ESD protection circuit structures (214, 252, and 256) and interface circuit 254, internal circuit 212, and/or pads (210, 216, 250, 264, and 266). Wirings 840 may be in the form of vertical vias that are formed through insulator layer 422 and/or region 820. Another region 830 may include internal circuit 212 and interface circuit 254, as well as wirings 850, and pads (210, 216, 250, 264, and 266).

Internal circuit 212 and ESD protection circuit structure 214 may each be electrically connected to pad (210 and 216). Internal circuit 212 may receive an input signal at an input terminal 218 and may provide an output signal at an output terminal 220. Pad 210 may receive an external power supply potential, such as VDD and pad 216 may receive an external reference potential such as VSS. In other embodiments, internal circuit 212 may be an internal power supply generator and may receive an external power supply potential at pad 216 and may provide an internal power supply potential to be used by internal circuits.

Pad 250 may receive an externally provided supply potential (for example VDD). Pad 264 may receive an externally provided power supply potential (for example VSS), and pad 266 may provide and/or receive an external signal (for example, a data or control signal).

ESD protection circuit structures (214, 252, and 256) in region 810 may be formed using planar IGFETs, n-type and/or p-type diffusion regions, and silicon control rectifiers (SCR), for example. Region 820 may include passive elements, such as polysilicon and/or metal resistors, incorporated in ESD structures (214, 252, and 256).

Internal circuit 212 and interface circuit 254 in region 830 may include p-type and n-type IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above substrate 402. Region 830 may generally have circuitry comprising p-type and n-type IGFETs having a plu-

ality of horizontally disposed channels that can be vertically aligned above substrate 402. Region 810 may include planar IGFETs or FinFETs fabricated using older technologies with more relaxed critical dimensions. In this way, reliable ESD protection circuit structures can be made more cheaply. Another advantage is that the region 830 exclusively has the normal operating circuits (i.e. exclusive of ESD structures which only operate when there is an ESD event). For example, if semiconductor device 800 is a microprocessor, the central processing unit (CPU), bus, and memory would all be located in region 830 and manufactured with a cutting edge state of the art process having smaller critical dimensions. By forming ESD protection circuit structures (214, 252, and 256) below the functional circuits, chip size can be reduced.

FIGS. 9 to 11 illustrate various ESD protection circuit structures that may be formed in regions (710 and 810) of FIGS. 7 and 8.

Referring now to FIG. 9, a circuit schematic diagram of an ESD protection circuit structure according to an embodiment is set forth and given the general reference character 900. ESD structure 900 may be a silicon controlled rectifier (SCR).

ESD structure 900 may include bipolar transistors (Q1 and Q2) and resistors (R910 and R920). Bipolar transistor Q1 may have an emitter terminal connected to a terminal 910, a base terminal commonly connected to a first terminal of resistor R920 and a collector terminal of bipolar transistor Q2, and a collector terminal commonly connected to a base terminal of bipolar transistor Q2 and a first terminal of resistor R910. Bipolar transistor Q2 may have an emitter terminal connected to a terminal 920. Resistor R910 may have a second terminal connected to terminal 920. Resistor R920 may have a second terminal connected to terminal 910.

ESD structure 900 may be used as ESD protection circuit structures (160, 170, 214, 252, and/or 256). When used as ESD protection circuit structure (140, 214, or 252) terminal 910 may be electrically connected to pads (110, 210, or 250) respectively, and terminal 920 may be electrically connected to pads (120, 216, or 264) respectively. When ESD protection circuit structure 900 is used as ESD protection circuit structure (170 or 256), terminal 910 may be connected to pads (130 or 266) respectively and terminal 920 may be electrically connected to pads (120 or 264) respectively.

Referring now to FIG. 10, a circuit schematic diagram of an ESD protection circuit structure according to an embodiment is set forth and given the general reference character 1000.

ESD protection circuit structure 1000 can include diodes (D1002 and D1004). Diode D1002 may have a cathode terminal connected to terminal 1020 and an anode terminal connected to terminal 1010. Diode D1004 may have a cathode terminal connected to terminal 1010 and an anode terminal connected to terminal 1030.

ESD protection circuit structure 1000 may be used as ESD protection circuit structures (170 and 256). When used as ESD protection circuit structures (170 or 256), terminal 1020 may be electrically connected to pads (110 or 250), respectively, terminal 1010 may be electrically connected to pads (130 or 266), respectively, and terminal 1030 may be electrically connected to pads (120 or 264), respectively.

Referring now to FIG. 11, a circuit schematic diagram of an ESD protection circuit structure according to an embodiment is set forth and given the general reference character 1100.

ESD structure **1100** can include IGFETs (**P1102** and **N1102**). IGFET **P1102** may have a source terminal and gate terminal commonly electrically connected to terminal **1120** and drain terminal electrically connected to terminal **1110**. IGFET **N1102** may have a source terminal and gate terminal commonly electrically connected to terminal **1130** and drain terminal electrically connected to terminal **1110**. IGFET **P1102** may be a p-type IGFET and IGFET **N1102** may be a n-type IGFET.

ESD structure **1100** may be used as ESD protection circuit structures (**170** and **256**). When used as ESD protection circuit structures (**170** or **256**), terminal **1120** may be electrically connected to pads (**110** or **250**), respectively, terminal **1110** may be electrically connected to pads (**130** or **266**), respectively, and terminal **1130** may be electrically connected to pads (**120** or **264**), respectively.

Referring now to FIG. **12**, a current-voltage diagram of an ESD protection circuit structure is set forth.

FIG. **12** is a current-voltage diagram of a typical ESD protection circuit structure. For example, FIG. **12** may be a current-voltage (I-V) diagram of SCR **900** of FIG. **9**.

For example, the current voltage diagram of FIG. **12** shows the SCR **900** in a forward blocking region **1202** in which there is minimal leakage current, which occurs when there is no ESD event. Once an ESD event occurs and the voltage spikes above a trigger voltage $V_{trigger}$ shown at point **1204** in the I-V diagram of FIG. **12**, the SCR **900** snaps back through snap back region **1206** toward a minimum holding voltage $V_{holding}$ at point **1208**. Then in the holding region **1210**, the SCR functions as a near ideal switch, the slope in holding region **1210** represents the on resistance of the SCR **1200**. This slope is proportional to the size of the SCR **1200**, thus a larger SCR **1200** dissipates more current at a lower holding voltage in the holding region. In designing the ESD protection circuit structures it is important to place the trigger voltage $V_{trigger}$ at a voltage that will be low enough that the IGFETs formed with vertically aligned and horizontally disposed channel regions in regions (**730** and **830**) will not breakdown during an ESD event.

Referring now to FIG. **13**, a circuit schematic diagram of an integrated circuit device according to an embodiment is set forth and given the general reference character **1300**.

Integrated circuit device **1300** can include an ESD protection circuit structure **1310** and an input buffer circuit **1320**. ESD protection circuit structure **1310** can be electrically connected to receive an input signal at a pad **1304**. The input signal may pass through the ESD protection circuit structure **1310** to terminal **1314**.

Input buffer circuit **1320** may receive the input signal from terminal **1314** and may provide an output signal at terminal **1308** (terminal **1308** may be an output terminal). Input buffer circuit **1320** may receive an enable signal **EN** at terminal **1322** and a reference potential V_{ref} at terminal **1324**.

ESD circuit structure **1310** can include an ESD protection circuit structure **1312** and a resistor **R1300**. ESD protection circuit structure **1312** may be electrically connected to pads (**1302**, **1304**, and **1306**). Pad **1302** may receive an externally provided power supply potential, such as **VDD**. Pad **1306** receive an externally provided power supply potential, such as **VSS**. Pad **1304** may receive an input signal, such as an address, data, and/or control signal, as just a few examples. ESD protection circuit structure **1312** may be electrically connected to a first terminal of resistor **R1300**. Resistor **R1300** may be electrically connected to terminal **1314**.

ESD structure **1312** may be an ESD protection circuit structure (**900**, **1000**, or **1100**), as just a few examples. In the

case of ESD structure (**900**, **1000**, or **1100**), pad **1304** may be electrically connected to terminal (**910**, **1010**, or **1110**), respectively.

Input buffer circuit **1320** may include IGFETs (**P1322**, **P1324**, **N1322**, **N1324**, and **N1326**). IGFET **P1322** may have a source terminal electrically connected to pad **1302** and commonly coupled to a source terminal of IGFET **P1324**. IGFET **P1322** may have a gate terminal and a drain terminal commonly connected to a gate terminal of IGFET **P1324** and a drain terminal of IGFET **N1322**. IGFET **P1324** may have a drain terminal connected to terminal **1308**. IGFET **N1322** may have a gate terminal coupled to receive a signal at terminal **1314** through ESD circuit structure **1310**. IGFET **N1322** may have a source terminal commonly connected to a source terminal of IGFET **N1324** and a drain terminal of IGFET **N1326**. IGFET **N1324** may have a drain terminal connected to terminal **1308** and a gate terminal connected to receive a reference potential V_{ref} at terminal **1324**. IGFET **N1326** may have a gate terminal connected to receive an enable signal **EN** at terminal **1322** and a source terminal connected to pad **1306**. Input buffer circuit **1320** may operate as a differential input buffer that is enabled when enable signal **EN** is at a logic high level and disabled when enable signal **EN** is at a logic low level.

IGFETs (**P1322**, **P1324**, **N1322**, **N1324**, and **N1326**) may each include a control gate that may surround a plurality of horizontally disposed channel regions that can be vertically aligned above a substrate as set forth in FIGS. **3A**, **3B**, **4**, **5**, and **6** and may be formed in region (**702** or **802**) as set forth in FIGS. **7** and **8**, respectively. ESD structure **1312** may be formed in regions **402** in FIGS. **7** and **8**, for example. Resistor **R1300** may be formed in region (**402** and/or **422**) in FIGS. **7** and **8**, for example. Resistor **R1300** may even be formed in region (**702** or **802**) as set forth in FIGS. **7** and **8**, respectively. Resistor **R1300** may be formed, for example, as a diffusion layer in region **402**, a metal layer in region **422**, and/or a metal layer in region (**702** or **802**) as set forth in FIGS. **7** and **8**, respectively.

Reference potential V_{ref} may provide a threshold voltage for determining the logic level of an input signal received at pad **1304**. For example, if the potential of the input signal received at pad **1304** is greater than reference potential V_{ref} , input buffer circuit **1320** may provide a logic high output at output terminal **1308**. However, if the potential of the input signal received at pad **1304** is less than reference potential V_{ref} , input buffer circuit **1320** may provide a logic low output at output terminal **1308**.

Referring now to FIG. **14**, a circuit schematic diagram of an integrated circuit device according to an embodiment is set forth and given the general reference character **1400**.

Integrated circuit device **1400** can include an ESD protection circuit structure **1410** and an output buffer circuit **1420**. ESD circuit structure **1410** can be electrically connected to a pad **1404** where an output signal may be provided externally to the integrated circuit device **1400**. The output signal may pass from output buffer **1420** through the ESD circuit structure **1410** to pad **1404**.

Output buffer circuit **1420** may receive an input signal from terminal **1408** and may provide an output signal at terminal **1414**.

ESD circuit structure **1420** can include an ESD structure **1412** and a resistor **R1400**. ESD structure **1412** may be electrically connected to pads (**1402**, **1404**, and **1406**). Pad **1402** may receive an externally provided power supply potential, such as **VDD**. Pad **1406** receive an externally provided power supply potential, such as **VSS**. Pad **1404** may receive a signal to be provided externally from inte-

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grated circuit device **1400**. ESD protection circuit structure **1412** may be electrically connected to a first terminal of resistor **R1400**. A second terminal of resistor **R1400** may be electrically connected to terminal **1414**.

ESD protection circuit structure **1412** may be an ESD protection circuit structure (**900**, **1000**, or **1100**), as just a few examples. In the case of ESD protection circuit structure (**900**, **1000**, or **1100**), pad **1404** may be electrically connected to terminal (**910**, **1010**, or **1110**), respectively.

Output buffer circuit **1420** may include IGFETs (**P1422** and **N1422**). IGFET **P1422** may have a source terminal electrically connected to pad **1402**. IGFET **P1422** may have a gate terminal and input terminal **1408** and a gate of IGFET **N1422**. IGFET **P1422** may have a drain commonly connected to a drain of IGFET **N1422** and a second terminal of resistor **R1400** at node **1414**. IGFET **N1422** may have a source terminal electrically connected to pad **1406**. Output buffer circuit **1420** may operate as an inverter logic circuit that provides current drive to a signal, such as a data signal or the like that is to be driven to components external to integrated circuit device **1400**.

IGFETs (**P1422** and **N1422**) may each include a control gate that may surround a plurality of horizontally disposed channel regions that can be vertically aligned above a substrate as set forth in FIGS. **3A**, **3B**, **4**, **5**, and **6** and may be formed in region (**702** or **802**) as set forth in FIGS. **7** and **8**, respectively. ESD protection circuit structure **1412** may be formed in regions **402** in FIGS. **7** and **8**, for example. Resistor **R1400** may be formed in region (**402** and/or **422**) in FIGS. **7** and **8**, for example. Resistor **R1400** may even be formed in region (**702** or **802**) as set forth in FIGS. **7** and **8**, respectively. Resistor **R1300** may be formed, for example, as a diffusion layer in region **402**, a metal layer in region **422**, and/or a metal layer in region (**702** or **802**) as set forth in FIGS. **7** and **8**, respectively.

Referring now to FIG. **15**, a circuit schematic diagram of an integrated circuit device according to an embodiment is set forth and given the general reference character **1500**. Integrated circuit device **1500** may include like constituents as integrated circuit device **1400** and such constituents may be designated by the same reference character and for brevity will not be discussed. Integrated circuit device **1500** may differ from integrated circuit device **1400** of FIG. **14**, in that an output buffer **1520** may have IGFETs (**P1522** and **N1522**) that are formed in substrate **402** and may be planar IGFETs or FinFETs, while other circuits, such as an internal circuit (**140** or **212**) of FIGS. **1** and **2**, respectively, may be formed from IGFETs that include a control gate that may surround a plurality of horizontally disposed channel regions that can be vertically aligned above a substrate as set forth in FIGS. **3A**, **3B**, **4**, **5**, and **6** and may be formed in region (**702** or **802**) as set forth in FIGS. **7** and **8**, respectively.

Integrated circuit device **1300** and integrated circuit devices (**1400** and **1500**) may be incorporated into integrated circuit devices (**100** and **200**). Integrated circuit device **1300** may have a separate pad electrically connected to the input terminal **1304** than a pad electrically connected to the output terminal **1404** of integrated circuit devices (**1400** and **1500**). Input buffer circuit **1320** and ESD structure **1310** of integrated circuit device **1300** may be used as interface circuit **150** and ESD circuit structure **170** of integrated circuit device **100** of FIG. **1**. Input buffer circuit **1320** and ESD structure **1310** of integrated circuit device **1300** may be used as interface circuit **254** and ESD circuit structure **256** of integrated circuit device **200** of FIG. **2**. Output buffer circuit **1420** and ESD structure **1410** of integrated circuit device **1400** may be used as interface circuit **150** and ESD circuit

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structure **170** of integrated circuit device **100** of FIG. **1**. Output buffer circuit **1420** and ESD structure **1410** of integrated circuit device **1400** may be used as interface circuit **254** and ESD circuit structure **256** of integrated circuit device **200** of FIG. **2**. Output buffer circuit **1520** and ESD structure **1510** of integrated circuit device **1500** may be used as interface circuit **150** and ESD circuit structure **170** of integrated circuit device **100** of FIG. **1**. Output buffer circuit **1520** and ESD structure **1510** of integrated circuit device **1500** may be used as interface circuit **254** and ESD circuit structure **256** of integrated circuit device **200** of FIG. **2**.

Referring now to FIG. **16**, a schematic diagram of an integrated circuit device according to an embodiment is set forth and given the general reference character **1600**.

Integrated circuit device **1600** may be like integrated circuit device **700** of FIG. **7**, except integrated circuit device **1600** may include a resistor **1610** formed in region **730** along with the circuitry comprising p-type and n-type IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above semiconductor substrate **402**.

Resistor **1610** may have one terminal electrically connected to pad **130** as well as ESD protection circuit structure **170** and another terminal electrically connected to interface circuit **150**. Resistor **1610** may correspond to resistors **R1400** in integrated circuit devices (**1400** and **1500**) as set forth in FIGS. **14** and **15**. Resistor **1610** may comprise a metal, such as copper, tungsten, aluminum, and/or titanium or even polysilicon, as just a few examples.

Referring now to FIG. **17**, a schematic diagram of an integrated circuit device according to an embodiment is set forth and given the general reference character **1700**.

Integrated circuit device **1700** may be like integrated circuit device **700** of FIG. **7**, except integrated circuit device **1700** may include a resistor **1710** formed in region **720** along with wirings **740**.

Resistor **1710** may have one terminal electrically connected to pad **130** as well as ESD protection circuit structure **170** and another terminal electrically connected to interface circuit **150**. Resistor **1710** may correspond to resistors **R1400** in integrated circuit devices (**1400** and **1500**) as set forth in FIGS. **14** and **15**. Resistor **1710** may comprise a metal, such as copper, tungsten, aluminum, and/or titanium or even polysilicon, as just a few examples.

Referring now to FIG. **18**, a schematic diagram of an integrated circuit device according to an embodiment is set forth and given the general reference character **1800**.

Integrated circuit device **1800** may be like integrated circuit device **700** of FIG. **7**, except integrated circuit device **1800** may include a resistor **1810** formed in region **710** along with ESD structures (**160** and **170**) and planar IGFETs fabricated using older technologies with more relaxed critical dimensions.

Resistor **1810** may have one terminal electrically connected to pad **130** as well as ESD protection circuit structure **170** and another terminal electrically connected to interface circuit **150**. Resistor **1810** may correspond to resistors **R1400** in integrated circuit devices (**1400** and **1500**) as set forth in FIGS. **14** and **15**. Resistor **1810** may comprise a metal, such as copper, tungsten, aluminum, and/or titanium, or polysilicon or a diffusion layer, as just a few examples.

Resistors (**R1300**, **R1400**, **1610**, **1710**, and/or **1810**) need sufficient resistance to provide a voltage drop between the pad **1404** and the interface circuit **150**. Resistors (**R1300** and **R1400**) may be about **1 kΩ** to **10 kΩ**.

The process minimum feature size of region **730** may be the control gate length of p-type and n-type IGFETs having

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a plurality of horizontally disposed channels that can be vertically aligned above semiconductor substrate **402**. Gate length is illustrated with reference to FIG. 6, in which "L" is the gate length of IGFET **410A/B**. In the embodiments, the minimum gate length may be about 5 nm or less.

The process minimum feature size of region **402** may be substantially greater. For example, a gate length of planar IGFETs formed in region **402** may be 10 nm or greater. An example of a planar IGFET is illustrated in FIG. 19.

Referring now to FIGS. 19, a cross-sectional schematic diagram of a planar IGFET that can be formed in region **402** is set forth and given the general reference character **1900**.

A planar IGFET formed in region **402** can include a semiconductor substrate **1902** in which source/drain regions **1918** may be formed, a gate insulating layer **1920**, a control gate **1914** and an insulating layer **1930**. Region **402** can include p-type IGFETs and n-type IGFETs. For example, an n-type IGFET may be formed by implanting n-type impurities into source/drain regions **1918** of a p-type semiconductor substrate **1902**. A p-type IGFET may be formed by providing a n-type well in semiconductor substrate **1902** and implanting p-type impurities into source/drain regions **1918**. Planar IGFET may have a gate length L1 of about 10 nm or greater. In this way, cost may be reduced as compared to the fabrication to IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure as in region **702**.

The IGFET formed in region **402** can be used as IGFETs (**P1522** and **N1522**) that are formed in substrate **402** as illustrated in FIG. 15, while other circuits, such as an internal circuit (**140** or **212**) of FIGS. 1 and 2, respectively, may be formed from IGFETs that include a control gate that may surround a plurality of horizontally disposed channel regions that can be vertically aligned above a substrate as set forth in FIGS. 3A, 3B, 4, 5, and 6 and may be formed in region (**702** or **802**) as set forth in FIGS. 7 and 8, respectively.

Referring now to FIGS. 20A and 20B, a cross-sectional schematic diagrams of a Fin field effect transistor (FinFET) type IGFET (i.e. FinFET) that can be formed in region **402** is set forth and given the general reference character **2000**.

FIG. 20A may be a cross-sectional schematic diagram of a FinFET along the width of a channel region **2016** and FIG. 20B may be a cross-sectional schematic diagram of a FinFET along the length of a channel region **2016** and between source/drain regions **2018**.

A FinFET formed in region **402** can include a semiconductor substrate **2002** in which source/drain regions **2018** may be formed, a gate insulating layer **2020**, a control gate **2014** and an insulating layer **2030**. Region **402** can include p-type FinFETs and n-type FinFETs. For example, an n-type FinFET may be formed by implanting n-type impurities into source/drain regions **2018** of a p-type semiconductor substrate **2002**. A p-type FinFET may be formed by providing a n-type well in semiconductor substrate **2002** and implanting p-type impurities into source/drain regions **2018**. FinFET may have a gate length L2 of about 7 nm or greater. In this way, cost may be reduced as compared to the fabrication to IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure as in region **702**.

The FinFET formed in region **402** can be used as IGFETs (**P1522** and **N1522**) that are formed in substrate **402** as illustrated in FIG. 15, while other circuits, such as an internal circuit (**140** or **212**) of FIGS. 1 and 2, respectively,

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may be formed from IGFETs that include a control gate that may surround a plurality of horizontally disposed channel regions that can be vertically aligned above a substrate as set forth in FIGS. 3A, 3B, 4, 5, and 6 and may be formed in region (**702** or **802**) as set forth in FIGS. 7 and 8, respectively.

Referring now to FIG. 21, a schematic diagram of an integrated circuit device having an ESD protection circuit structure having a plurality of horizontally current carrying regions that can be vertically aligned above a substrate is set forth and given the general reference character **2100**.

Integrated circuit device includes regions (**710**, **720**, and **730**). As noted earlier, region **710** may be formed with a technology node that is older and cheaper than region **730**. Region **710** may include planar IGFETs and ESD structures. However, integrated circuit device **2100** may differ in that an ESD structure may be formed in region **730** and may include diodes (**D2102** and **D2104**). Diodes (**D2102** and **D2104**) may include a plurality of horizontally disposed current carrying regions that can be vertically aligned above a substrate region **402**. The current carrying regions may include a first impurity doped region **2112** and a second impurity doped region **2114**. Each diode (**D2102** and **D2104**) may include a cathode terminal **2116** and an anode terminal **2118**. The anode terminal **2118** of diode **D2102** may be electrically connected to the cathode terminal **2116** of diode **D2104** and may be electrically connected to a pad **2110** which may be electrically connected to provide or receive an external signal. The cathode terminal **2116** of diode **D2102** may be electrically connected to a pad **2120**. Pad **2120** may receive an externally provided power supply potential, such as VDD. The anode terminal **2118** of diode **D2104** may be electrically connected to a pad **2130**. Pad **2130** may receive an externally provided power supply potential, such as VSS.

The ESD protection circuit structure of FIG. 21 including diodes (**D2102** and **D2104**) may correspond to ESD protection circuit structure **1000** of FIG. 10 and may be used accordingly. Pad **2130** may correspond to terminal **1030**, pad **2110** may correspond to terminal **1010**, and pad **2120** may correspond to terminal **1020**. Likewise, diode **D2102** may correspond to diode **D1002** and diode **D2104** may correspond to diode **D1004**.

Diodes (**D2102** and **D2104**) may be formed by forming a layered crystal of two materials over dielectric region **422**. For example, layers of silicon and silicon germanium may be formed. The silicon layer may form the first and second impurity doped regions (**2112** and **2114**), i.e. the current carrying regions. After a vertical etch, the silicon germanium layers may be etched by using a chemical that can selectively etch silicon germanium with the cathode and anode terminals (**2116** and **2118**) used as support structures. Next, a dielectric layer **2122** (may be formed using atomic layer deposition of a dielectric, for example, silicon dioxide. The first impurity doped region **2112** may be doped with n-type carriers, such as phosphorous and/or arsenic, for example. The second impurity doped region **2114** may be doped with p-type carriers, such as boron, for example. The doping may be done by implantation with a mask layer over regions other than the desired regions to receive the impurities. In this way each of the plurality of horizontally disposed current carrying regions may form a p-n junction diode in parallel with each other.

Diodes (**D2102** and **D2104**) may be formed in conjunction with insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can

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be vertically aligned above a substrate with each channel being surrounded by a gate structure as discussed above.

Referring now to FIG. 22, a diagram of an integrated circuit device according to an embodiment is set forth and given the general reference character 2100.

Integrated circuit device 2200 may differ from integrated circuit devices of previous embodiments in that a region 2210 may be disposed between regions (720 and 730), otherwise integrated circuit device 2200 may be substantially the same as previous embodiments. Region 2210 may be a crystalline semiconductor layer. For example, region 2210 may be silicon material. Region 2210 may be silicon, silicon carbide, epitaxial silicon, as just a few examples. Region 2210 may improve the manufacturability of layers used to form the horizontally disposed and vertically aligned channel regions. Integrated circuit device 2200 may include insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure formed in region 730 as discussed above, as well as ESD protection circuit structures formed in regions (710, 720, and 730) as discussed in previous embodiments.

Referring now to FIG. 23, a circuit schematic diagram of an internal circuit and an ESD protection circuit structure according to an embodiment is set forth and given the general reference character 2300.

Circuit 2300 can include an internal circuit 2310 and an ESD protection circuit structure 2320.

Internal circuit 2300 may receive a power supply potential from a pad 2302. The power supply potential from pad 2302 may be an externally applied power supply potential such as VDD. Internal circuit 2300 may receive a power supply potential from a pad 2306. The power supply potential from pad 2306 may be an externally supplied power supply potential such as VSS. Internal circuit may receive an input signal from an input terminal 2308 and provide an output signal at a terminal 2314. Internal circuit 2310 may include a p-type IGFET P2312 and an n-type IGFET N2312. Both p-type IGFET P2312 and n-type IGFET N2312 may each include a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure.

P-type IGFET P2312 may have a source terminal electrically connected to pad 2302. N-type IGFET N2312 may have a source terminal electrically connected to pad 2306. P-type IGFET P2312 and N-type IGFET N2312 may have gate terminals commonly connected to receive the input signal from input terminal 2308 and drain terminals commonly connected to provide the output signal at output terminal 2314.

Internal circuit 2310 and may be used as internal circuit 140 and/or internal circuit 212 of FIGS. 1 and 2, respectively. ESD protection circuit structure 2320 may be used as ESD protection circuit structure 160 or any/each or ESD circuit structures (214 and 252) of FIGS. 1 and 2, respectively.

Internal circuit 2310 may not be electrically connected to receive or provide a signal external to the integrated circuit device.

ESD protection circuit structure 2320 may include two ESD protection circuits, a diode D2324 and an ESD protection circuit 2322, each electrically connected between pads (2302 and 2306). In this way, ESD protection circuit structure 2320 may provide protection for an ESD event at either pad (2302 or 2306), that receive externally provided power supply potentials. ESD protection circuit 2322 may

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be a SCR such as SCR 900 illustrated in FIG. 9. ESD protection circuit 2322 can be provided in regions (710 or 810),

Diode D2324 can have a cathode terminal electrically connected to pad 2302 and an anode terminal electrically connected to pad 2306. Diode D2324 can be formed in regions (710 or 810) as a p-n junction or in region (730 or 830). When diode D2324 is formed in regions (730 or 830), diode D2324 may include a plurality of horizontally current carrying regions that can be vertically aligned above a substrate as illustrated with respect to diodes (D2104 and D2102) in FIG. 21.

Referring now to FIG. 24, an integrated circuit device including a circuit having an ESD protection circuit structure according to an embodiment is set forth in a circuit schematic diagram and given the general reference character 2400. Integrated circuit device 2400 may have similar circuit constituents as integrated circuit device 200 of FIG. 2 and such constituents may have the same reference character. Integrated circuit device 2400 may include a first circuit section 202, a second circuit section 2410, and a third circuit section 2440. First circuit section 202 may include circuits that only have external connections to a power supply potential and/or a ground (VSS) potential. Second circuit section 2410 may include circuits that have external connections to a power supply potential, a ground potential, and/or a pad coupled to receive an external signal, such as a data signal, control signal or a clock signal, as just a few examples. Third circuit section 2440 may include circuits that have external connections to a power supply potential, a ground potential, and/or a pad coupled to provide an external signal, such as a data signal, control signal or a clock signal, as just a few examples.

First circuit section 202 may include an internal circuit 212 and an ESD protection circuit structure 214. Internal circuit 212 and ESD structure 214 may each be electrically connected to pad (210 and 216). Internal circuit 212 may receive an input signal at an input terminal 218 and may provide an output signal at an output terminal 220. Pad 210 may receive an external power supply potential, such as VDD and pad 216 may receive an external reference potential such as VSS. In other embodiments, internal circuit 212 may be an internal power supply generator and may receive an external power supply potential at pad 210 and may provide an internal power supply potential to be used by internal circuits. The input terminal 218 and the output terminal 220 of internal circuit 212 are not electrically connected to any pad that can receive or provide a signal external to the integrated circuit device 200.

Second circuit section 2410 may include pads (2412, 2414, and 2416), an ESD protection circuit structure 2422, an input buffer circuit 2420, and ESD protection circuit structure 2418. Input buffer circuit 2420 may receive an external signal at pad 2416 through ESD protection circuit structure 2418 and may provide an internal signal at terminal 2424. Input buffer circuit 2420 may be electrically connected to pads (2412 and 2414). Pads (2412 and 2414) may respectively receive an external power supply potential (such as VDD) and a reference potential (such as VSS). ESD structure 2422 may be electrically connected between pads (2412 and 2414). ESD structure 2418 may be electrically connected to pads (2412, 2414, and 2416).

Third circuit section 2440 may include pads (2442, 2444, and 2446), an ESD protection circuit structure 2448, an output buffer circuit 2450, and ESD protection circuit structure 2452. Output buffer circuit 2450 may receive an internal signal at terminal 2454 and may provide an external signal

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at pad **2446** through ESD protection circuit structure **2452**. Output buffer circuit **2450** may be electrically connected to pads (**2442** and **2444**). Pads (**2442** and **2444**) may respectively receive an external power supply potential (such as VDD) and a reference potential (such as VSS). ESD structure **2448** may be electrically connected between pads (**2442** and **2444**). ESD structure **2452** may be electrically connected to pads (**2442**, **2444**, and **2446**).

In one embodiment, internal circuit **212** may include insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure. Input buffer circuit **2420** may include insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure. Output buffer circuit **2452** may include insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure. Internal circuit **212**, input buffer circuit **2420**, and output buffer circuit **2450** may include p-type and n-type IGFETs. In one embodiment, ESD protection circuit structures (**214**, **2418**, **2422**, **2448**, and **2452**) may include electrical components (such as diodes, transistors, and/or resistors) formed with a plurality of horizontally disposed cathodes and anodes that can be vertically aligned above a substrate. In one embodiment, ESD protection circuit structures (**214**, **2418**, **2422**, **2448**, and **2452**) may include electrical components (such as diodes, transistors, SCRs and/or resistors) formed in the substrate. In one embodiment input buffer circuit **2420** and/or output buffer circuit **2452** may include electrical components (such as IGFETs) formed in the substrate.

When ESD protection circuit structures (**214**, **2418**, **2422**, **2448**, and **2452**) are formed in a semiconductor substrate of integrated circuit device **2400**, a process having larger critical dimensions (i.e. an older and cheaper process) may be used. The semiconductor substrate may then be sent to a state of the art fabrication facility to form the circuit including insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure as will be discussed further in the instant specification.

Integrated circuit device **1300** and integrated circuit devices (**1400** and **1500**) may be incorporated into integrated circuit device **2400**. Input buffer circuit **1320** and ESD structure **1310** of integrated circuit device **1300** may be used as input circuit **2420** and ESD circuit structure **2418** of integrated circuit device **2400** of FIG. **24**. Output buffer circuit **1420** and ESD structure **1410** of integrated circuit device **1400** may be used as output buffer circuit **2450** and ESD circuit structure **2452** of integrated circuit device **2400** of FIG. **24**. Output buffer circuit **1520** and ESD structure **1510** of integrated circuit device **1500** may be used as output buffer circuit **2450** and ESD circuit structure **2452** of integrated circuit device **2400** of FIG. **24**.

Power supply potentials externally provided to pads (**210**, **2412**, and **2442**) may be different power supply potentials, such as a first potential (VDD1) for internal circuit **212**, a second potential (VDD2) for input buffer circuit **2420**, and/or a third potential (VDD3) for output buffer circuit **2450**.

Input buffer **1300** of FIG. **13** and output buffers (**1400** and **1500**) of FIGS. **14** and **15** may be incorporated into integrated circuit devices **2400**.

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Integrated circuit devices (**700**, **800**, **1600**, **1700**, **1800**, **2100**, and **2200**) may be contiguous structures, such that, regions may be deposited or bonded in a semiconductor fabrication facility and preferably all formed on a contiguous wafer in a multiple of units and then separated before packaged or set in a multi-chip package. For example, regions (**710**, **720**, and **730**) may be contiguous regions with virtually no separation other than a region border formed by a change of materials. Bonding of regions may be performed using wafer to wafer bonding, for example region **710** may be formed on a first semiconductor wafer and regions (**720** and **730**) may be formed on a second semiconductor wafer, then the first and second wafer may be bonded using a wafer to wafer bonding technique followed by dicing and packaging to form the integrated circuit device. Alternatively, region **710** may be formed on a first semiconductor wafer and regions (**720** and **730**) may be formed on a second semiconductor wafer, then the either the first or second wafer may be diced and a die pick and place may be used to place dies on the first or second intact wafer, followed by dicing and packaging to form the integrated circuit device.

It is understood that the term pad may be any circuit connection that is electrically connected to provide or receive a signal or a potential externally to the integrated circuit device. Such a connection can be a conduit for an ESD event.

Electrically connected can be a connection through a wiring other passive component such as a resistor.

A voltage may be expressed as a potential.

A signal can be a data or control signal that can transition between logic levels, as just a few examples. A signal is not a power supply potential used to provide power to circuitry.

Other electrical apparatus other than semiconductor devices may benefit from the invention.

While various particular embodiments set forth herein have been described in detail, the present invention could be subject to various changes, substitutions, and alterations without departing from the spirit and scope of the invention. Accordingly, the present invention is intended to be limited only as defined by the appended claims.

What is claimed is:

1. An integrated circuit device, comprising:

a first region;

a second region formed over the first region;

an internal circuit coupled between a first power supply potential and a second power supply potential, the internal circuit comprising an input terminal coupled to receive an input signal and an output terminal coupled to provide an output signal, the internal circuit comprising a first insulated gate field effect transistor (IGFET) formed in the second region, the first IGFET comprising a plurality of channels that extend in a first direction and are substantially aligned in a second direction to that crosses the first direction, and a control gate that substantially surrounds the plurality of channels, wherein the plurality of channels provide a controllable impedance path between a first source/drain terminal and a second source/drain terminal; and

a first electrostatic discharge (ESD) protection circuit structure that is provided in the first region, and is electrically connected between the first power supply potential and the second power supply potential, wherein the control gate is coupled to receive the input signal and the first source/drain terminal is coupled to provide the output signal, and the internal circuit is not connected to receive a signal generated external to the

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- integrated circuit device and is not connected to provide a signal external to the integrated circuit device.
2. The integrated circuit device of claim 1, wherein: the first ESD protection circuit structure comprises a diode.
 3. The integrated circuit device of claim 2, wherein: the diode comprises an anode terminal electrically connected to the first power supply potential and a cathode region electrically connected to the second power supply potential.
 4. The integrated circuit device of claim 1, wherein: the first ESD protection circuit structure comprises a silicon controlled rectifier (SCR) circuit coupled between the first power supply potential and the second power supply potential.
 5. The integrated circuit device of claim 1, wherein: the internal circuit further comprises a second IGFET formed in the second region, the second IGFET comprising a plurality of channels that extend in the first direction and are substantially aligned in the second direction, and a control gate that substantially surrounds the plurality of channels, wherein the plurality of channels provide a controllable impedance path between a third source/drain terminal and a fourth source/drain terminal, and the third source/drain terminal is coupled to the first source/drain terminal; and the first IGFET is a p-type IGFET and the second IGFET is a n-type IGFET.
 6. The integrated circuit device of claim 5, wherein: the p-type IGFET and n-type IGFET are configured as an inverter circuit.
 7. The integrated circuit device of claim 1, wherein: the first and second power supply potentials are provided from a source that is external to the integrated circuit device.
 8. The integrated circuit device of claim 1, further comprising: an input buffer circuit coupled to receive an input signal provided from a source that is external to the integrated circuit device, the input signal received at a first pad.
 9. The integrated circuit device of claim 8, further comprising: a second ESD protection circuit structure having a first terminal electrically connected between the first pad and an input terminal of the input buffer circuit.
 10. The integrated circuit device of claim 9, wherein: the second ESD protection circuit structure comprises a resistor electrically connected between the first terminal of the second ESD protection circuit structure and the input terminal of the input buffer circuit.
 11. The integrated circuit device of claim 10, wherein: the second ESD protection circuit structure further comprises a first diode having an anode terminal coupled to the first terminal of the second ESD protection circuit structure.

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12. The integrated circuit device of claim 11, wherein: the second ESD protection circuit structure further comprises a second diode having a cathode terminal coupled to the first terminal of the second ESD protection circuit structure.
13. The integrated circuit device of claim 11, wherein: the first diode comprises a plurality of current carrying regions disposed substantially in the first direction.
14. The integrated circuit device of claim 1, further comprising: an output buffer circuit coupled to receive an input signal and provide an output signal externally to the integrated circuit device through a first pad.
15. The integrated circuit device of claim 14, further comprising: a second ESD protection circuit structure having a first terminal electrically connected between the first pad and an output terminal of the output buffer circuit.
16. The integrated circuit device of claim 15, wherein: the second ESD protection circuit structure further comprises a first diode having an anode terminal coupled to the first terminal of the second ESD protection circuit structure.
17. The integrated circuit device of claim 16, wherein: the second ESD protection circuit structure further comprises a second diode having a cathode terminal coupled to the first terminal of the second ESD protection circuit structure.
18. The integrated circuit device of claim 16, wherein: the first diode comprises a plurality of current carrying regions disposed substantially in the first direction.
19. The integrated circuit device of claim 1, wherein: the first ESD protection circuit structure is provided in a semiconductor substrate.
20. An integrated circuit device, comprising: a first region; a second region over the first region; an internal circuit coupled between a first power supply potential and a second power supply potential, the internal circuit comprising an input terminal configured to receive an input signal, an output terminal configured to provide an output signal, and a first insulated gate field effect transistor (IGFET), wherein the first IGFET is provided in the second region, comprises a plurality of channels and a control gate configured to receive the input signal, and is configured to provide a controllable impedance path between a first source/drain terminal and a second source/drain terminal; and a first electrostatic discharge (ESD) protection circuit structure that is provided in the first region, and is electrically connected between the first power supply potential and the second power supply potential.

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